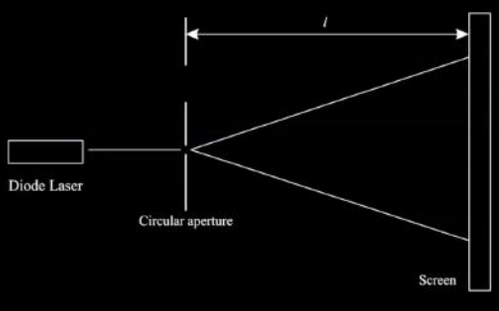
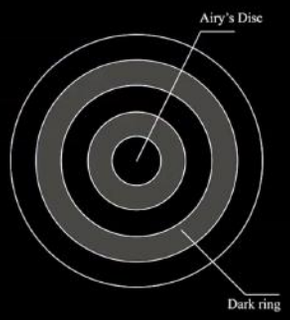
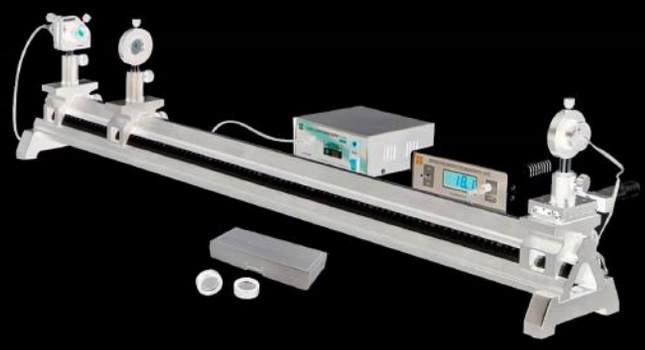
Failure Of Ray Optics: Diffraction:







If we pass a parallel beam of light through a circular hole and make it fall on screen, according to ray optics intuition, we expect to obtain a circular illumination with aperture equal to that of hole, however in practice with sufficiently small holes, we obtain a pattern as the one shown in adjacent figure. Such an observation would appear weird according to the logic of Geometrical optics thus an alternative theory of light was required.

(Besides diffraction there is also the phenomenon of interference which would be unsettling for a firm believer of Geometrical optics, (such as Young's double slit experiment).

ACTUAL DIFFRACTION IMAGE.

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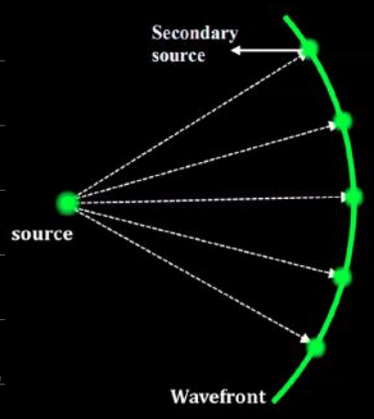
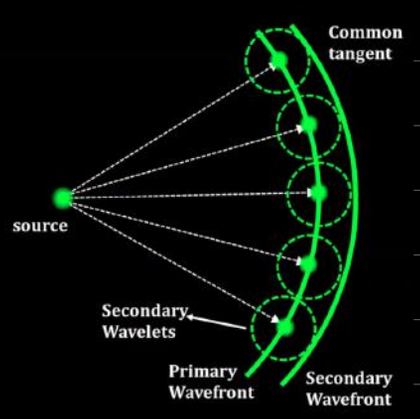
Huygen's Wave Theory:

To explain the newly observed phenomena of diffraction and interference of light, a new theory was required which was also capable of explaining previously known optical phenomena such as refraction and reflection.





A dutch physicist by the name Christiaan Huygen proposed a wave theory of light which could explain many of the phenomena satisfactorily at that time. The key ideas in this

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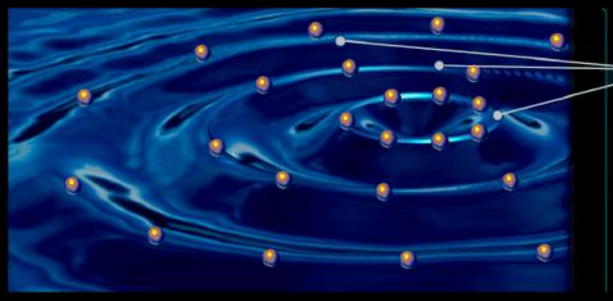
wave theory are as follows:



→ As light propagates through a medium, particles of the medium oscillate.

→ **WAVEFRONT**: A surface joining all particles oscillating in same phase is called a wavefront. For instance for ripples spreading on surface of water, the wavefronts are circular (see the figure above).

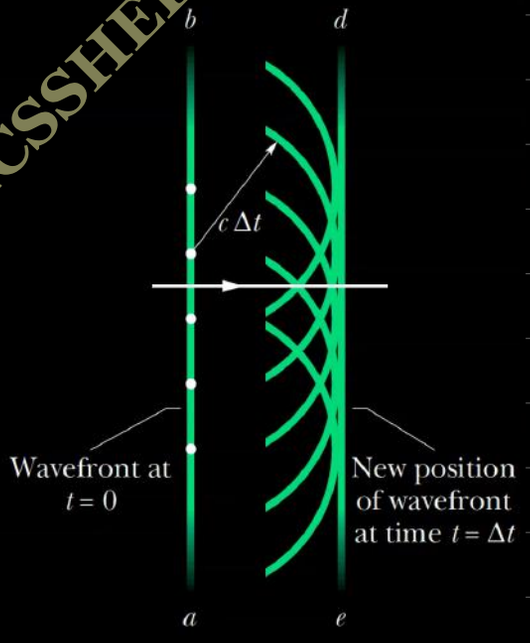
→ Every point on the wavefront becomes a source of secondary wavelets which travel in all directions with speed of light.

→ The tangent surface to these secondary wavelets gives us the new position of the wavefront.

→ Huygen's theory can be used to satisfactorily explain refraction, reflection, diffraction and interference of light.

→ Original Huygen's theory failed to explain quantitative intensity in the diffraction and interference patterns however it was modified by other physicists to justify the quantitative results.

→ Huygen's theory fails to explain photoelectric effect and polarization.



Rectilinear Propagation (Wavelets).

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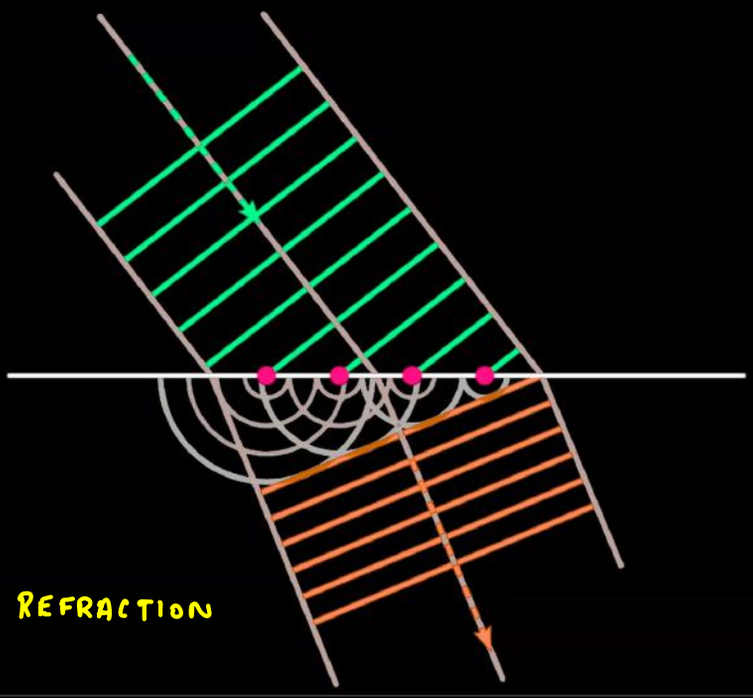
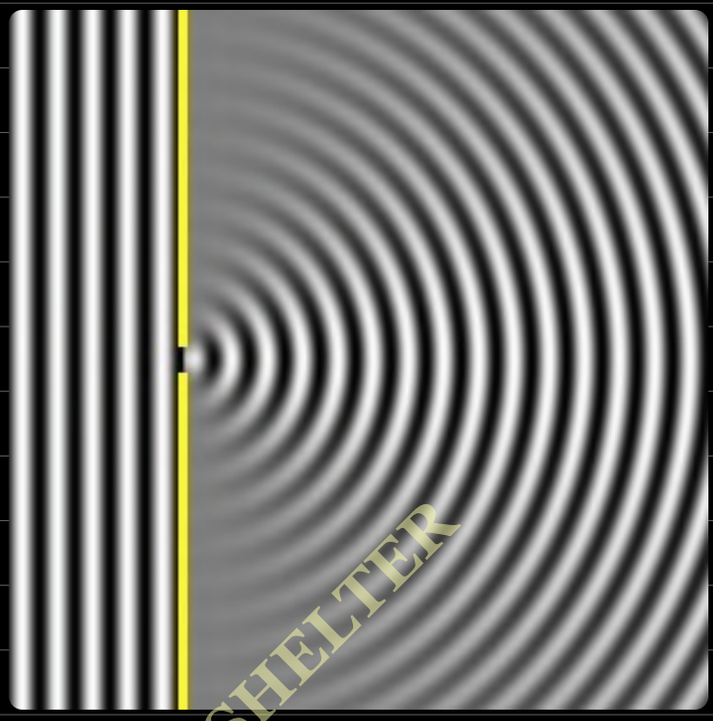
Explaining Diffraction And Refraction Using Huygen's Theory:

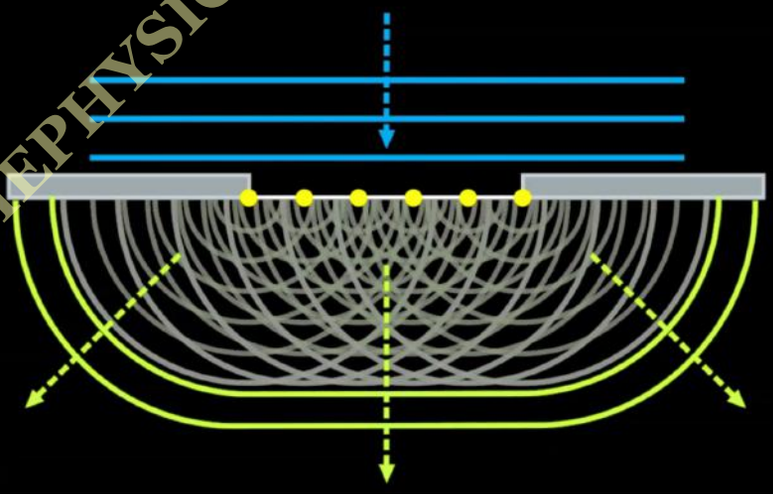


Refraction



Diffraction

The above and adjacent graphics give us a visual idea about how Huygen's principle explains the phenomena of refraction and diffraction.



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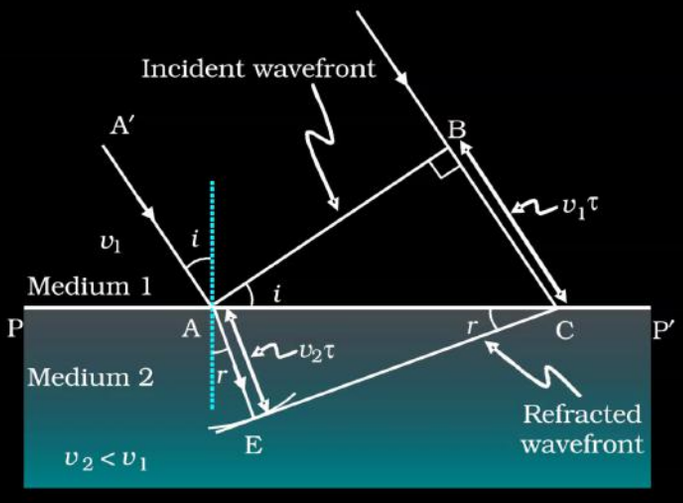
Refraction Of A Plane Wave Using Huygen's Principle: (NCERT):

Consider the adjacent figure. Here PP' is the boundary separating the two media, (Medium 1 and medium 2). v_1 and v_2 are speeds of light in the respective media. We assume a plane wavefront propagating along A'A incident on the interface at an angle i as shown. Let τ be the time taken by the wavefront to travel the distance BC.



Thus $BC = v_1 \tau$. To determine the shape of refracted wavefront, we draw a sphere of radius $v_2 \tau$ from point A in the second medium ($\because$ speed is v_2), and draw a tangent plane CE from point C on the sphere to obtain the refracted wavefront. Now:

$$\sin i = BC/AC = v_1 \tau / AC$$

$$\sin r = AE/AC = v_2 \tau / AC$$

Here i and r are angles of incidence and refraction respectively. Thus:

$$\frac{\sin i}{\sin r} = \frac{v_1}{v_2}$$, thus if $r < i$, the ray will bend towards the normal and speed of light is smaller in the second medium. This is in agreement with experiments (and this conclusion is just the opposite of Newton's corpuscular theory which incorrectly predicts $v_2 > v_1$ if $r > i$). Now by definition of refractive index:

$$n_1 = c/v_1 \text{ and } n_2 = c/v_2 \text{ thus:}$$

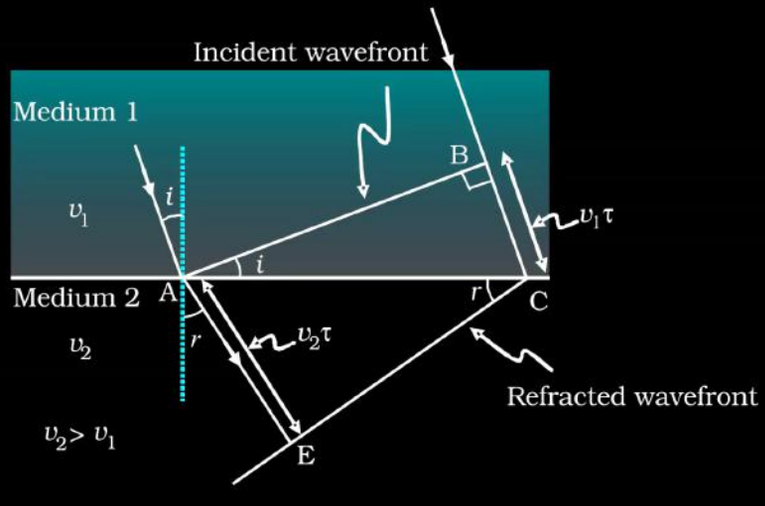
$$\frac{\sin i}{\sin r} = \frac{n_2}{n_1}$$ this is the well known snell's law QED.

Refraction From A Rarer Medium:

Just as we made the wavefront diagram for refraction from a denser medium, we can also draw a similar diagram for refraction from a rarer medium. Here since $v_2 > v_1$, it can be readily inferred that $r > i$.

($\because \sin r = v_2 \tau / AC$

$\sin i = v_1 \tau / AC$).



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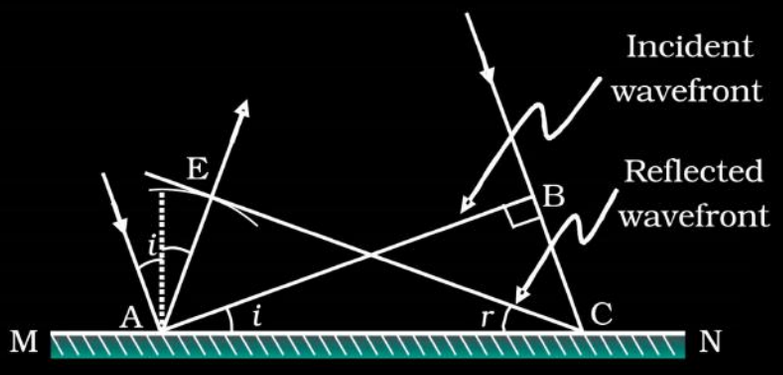
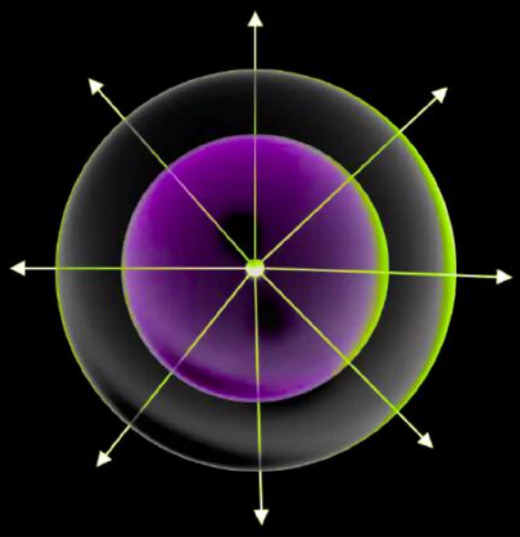
Reflection Of Plane Wave Using Huygen's Principle:



Let AB be the incident plane wavefront on the mirror and let v be the speed of the wave. Let t be the time taken by the wavefront to advance from point B to C then, $BC = vt$. To find the reflected wavefront, we draw a sphere of radius vt from point A and draw the tangent plane CE from point C to this sphere. Now obviously $AE = BC = vt$. The triangles EAC and BCA are congruent thus $i = r$. This is the law of reflection.

Some Standard Wavefronts And Their Properties:

Spherical Wavefronts: Such wavefronts are obtained from point or spherical isotropic sources. Here intensity varies with $1/r^2$ and hence the electric field amplitude declines with $1/r$ (so also magnetic field amplitude). Recall from chapter on EM waves that $\vec{I} = \vec{u}c$ and, $u_E = u_B = \frac{1}{2}\epsilon_0 E^2 = \frac{B^2}{2\mu_0}$. The $1/r^2$ variation of intensity simply follows from the fact that at a distance r, the power is spread out in to a sphere of radius r.



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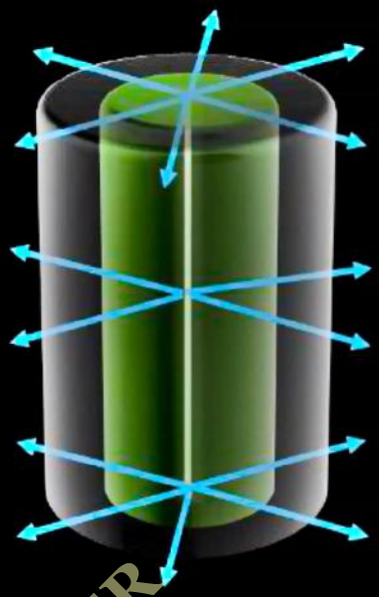
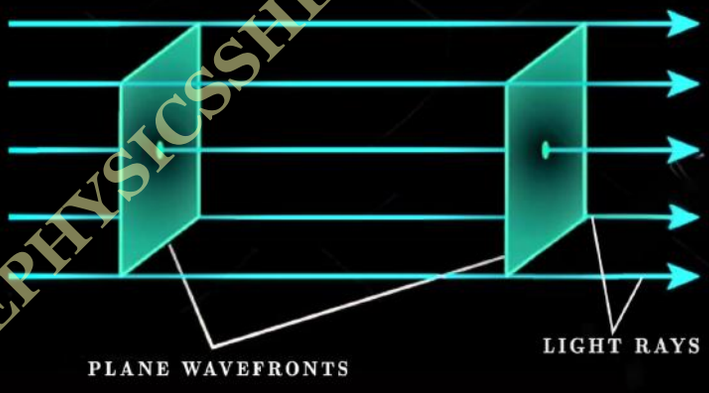
Cylindrical Wavefront : Cylindrical wavefronts are obtained from linear or cylindrical sources. Here since by the time wavefront attains a radius r, it spread over an area $2\pi rl$ thus intensity is proportional to $1/r$ and amplitude of electric field is thus proportional to $1/\sqrt{r}$.



Plane Wavefront : Plane wavefronts occur when there are point sources at infinity (for plane waves rays must be parallel to each other). At sufficiently large distances from the source, cylindrical and spherical wavefronts may be modelled as plane wavefronts for certain purposes. In plane wavefronts, Intensity remains constant since there is no spreading out of energy (cross section remains constant) thus amplitude also remains constant.



Summary Of Results :

Sr. No.	Wavefront	Shape of Light Source	Diagram or shape of wavefront	Variation of amplitude (A) with distance	Variation of Intensity (I) with distance
1.	Spherical	Point source		$A \propto \frac{1}{r}$	$I \propto \frac{1}{r^2}$
2.	Cylindrical	Linear source/ Slit		$A \propto \frac{1}{\sqrt{r}}$	$I \propto \frac{1}{r}$
3.	Plane	Extended Large source/ Point source at very large distance		A: constant $A \propto r^0$	I: constant $I \propto r^0$

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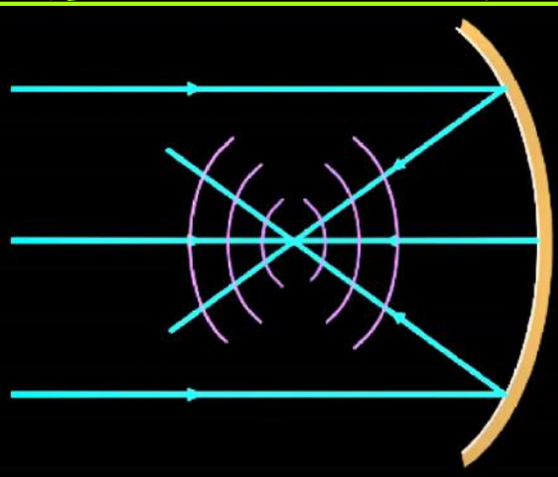
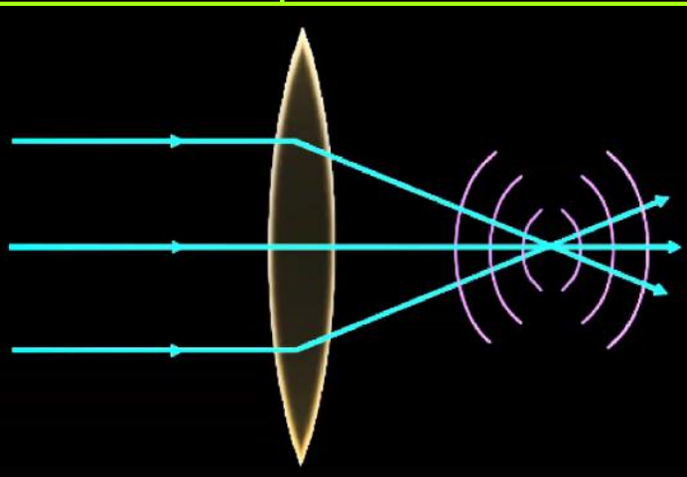
WAVEFRONT SHAPES DUE TO STANDARD REFLECTING / REFRACTING DEVICES:



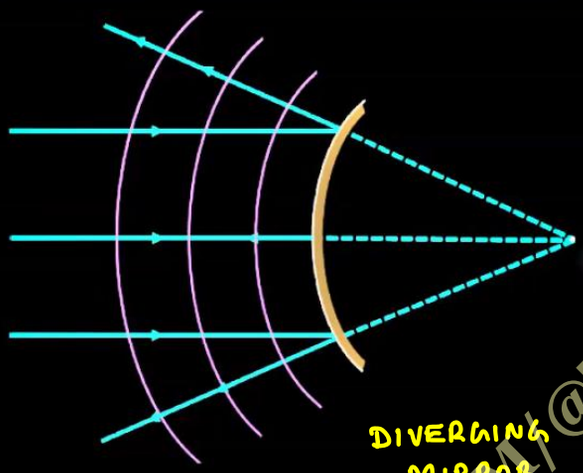
Spherical wavefront

CONVERGING MIRROR



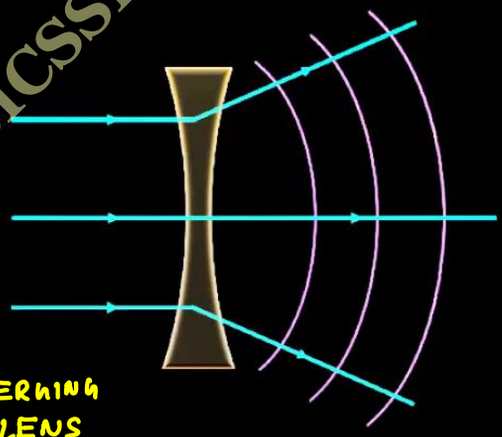
Spherical wavefront

CONVERGING LENS



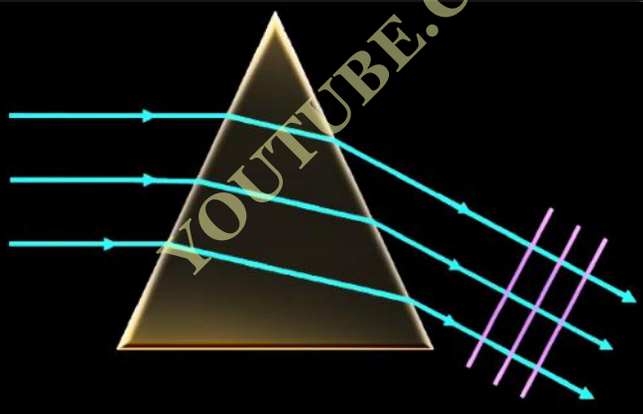
DIVERGING MIRROR

Spherical wavefront



DIVERGING LENS

Spherical wavefront



PRISM.

→ Wave fronts can easily be drawn using the ray diagrams. All wavefronts are perpendicular to the rays (just as equipotentials are perpendicular to the field lines).

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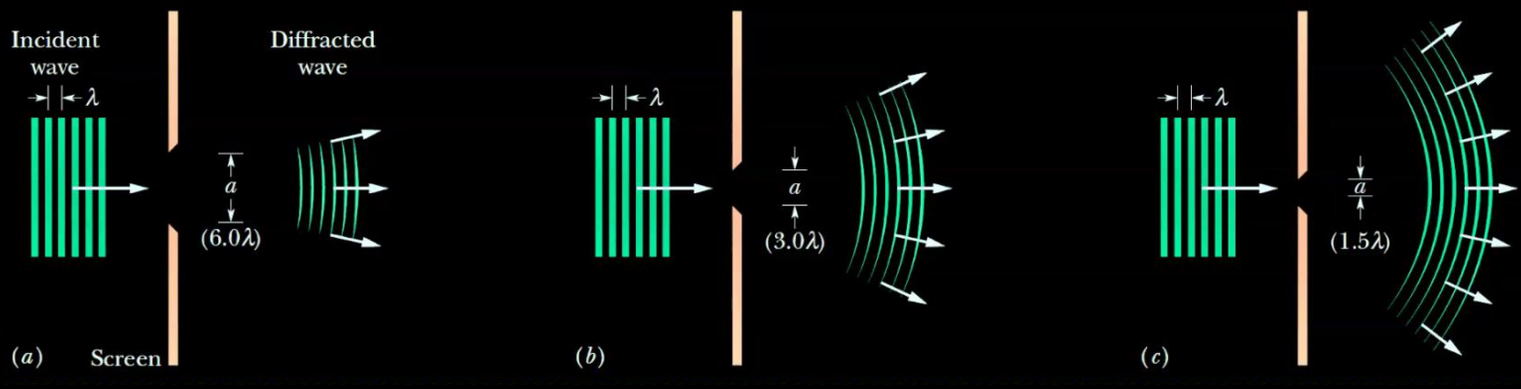
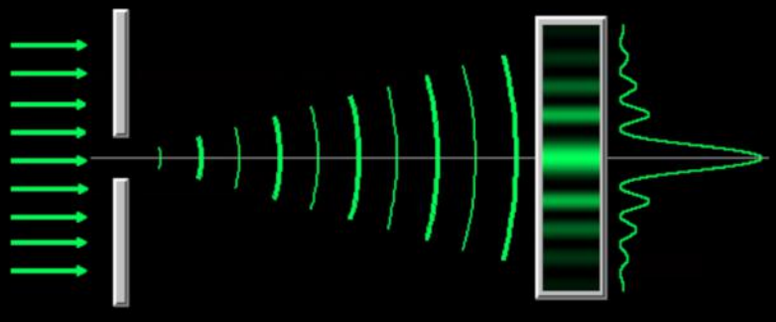
Slit Diffraction (Qualitative):



Whenever light is made to pass through apertures comparable to its wavelength, the diffraction effects become more and more significant. As can be seen in the figures above, as the aperture is decreased from 6λ to 1.5λ, the fanning out of light increases significantly. If we place a screen in front of the diffraction aperture, we get alternate bands of maxima and minima with practically all of the light concentrated in the central maximum.

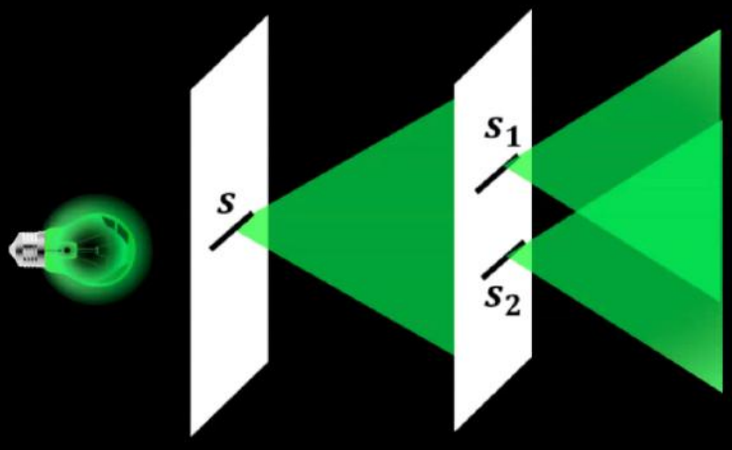
→ For a quantitative idea see the adjacent figure and see how the central band is very bright and other bands are progressively dimmer.

→ A mathematical expression for this intensity variation will be derived in a later section.



Coherent Sources:

Two or more sources are said to be coherent if they have same frequency and maintain constant phase difference. Coherent sources are obtained by splitting the light from same source in various ways. In general light waves from different sources are incoherent.



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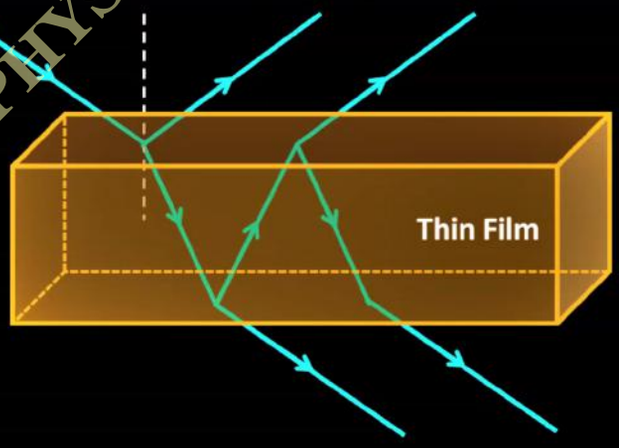
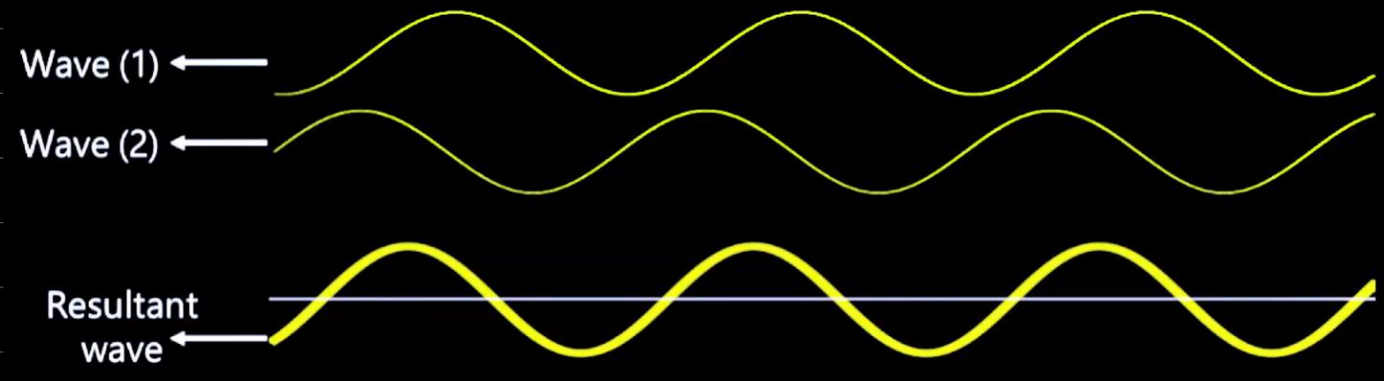
Obtaining Coherent Sources: Typically two distinct methods are used for creating coherent sources:

Division Of Wavefront: In this method a wavefront is either refracted, reflected or passed through different slits to form two distinct coherent sources. For instance placing a double-slit in path of slit diffraction wavefronts leads to formation of two coherent sources S_1 and S_2 which in turn send out secondary wavelets in all directions. (Here two distinct points on wavefront are involved)

Division Of Amplitude:
Here the amplitude of the incoming beam is divided in to two parts either by partial reflection or refraction. These refracted / reflected beams are then made to interfere with each other by suitable set up to produce bright and dark fringes.



Interference Of Light:



Just like mechanical waves, whenever two Electromagnetic waves

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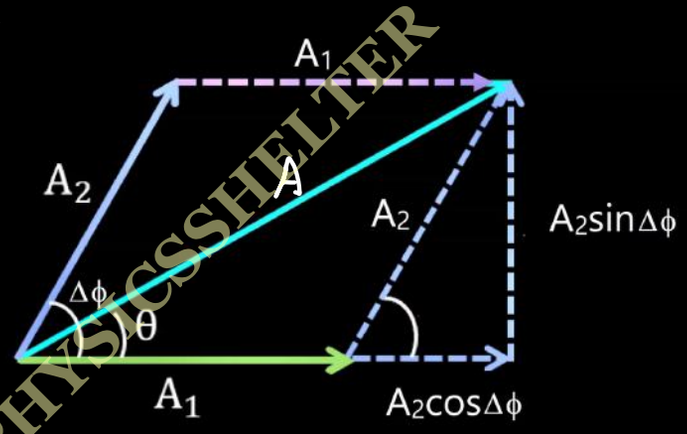
superpose with each other, redistribution of energy takes place. Recall that superposition of two simple harmonic motions of same frequency leads to a simple harmonic motion whose Amplitude depends on the phase difference and amplitudes of the superposing disturbances. In the similar manner amplitude of electric field oscillations due to superposition of two waves depends on the amplitudes and phase differences of individual electric field disturbances.

Recap Of Phasor Addition / Intensity Superposition From SHM / Waves :

Resultant Amplitude Of Phasors: If two phasors of Amplitudes A_1 and A_2 having a phase difference $\Delta\phi$ interfere, the net resultant amplitude of the phasor is given as:



$$A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\Delta\phi}$$

Refer to the adjacent figure. Here θ can be obtained using the relation:

$$\tan\theta = \frac{A_2\sin\Delta\phi}{A_1 + A_2\cos\Delta\phi}$$

Resultant Intensity Upon Superposition :

Since intensity is proportional to square of Electric field amplitude, thus resultant intensity of superposition of two Electromagnetic waves (Same plane of polarization) is given by:

$$I = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\Delta\phi$$

$$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$$

$$I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

Relation Between Phase Lag / Distance Lag / Time Lag :

Let x_1 and x_2 be two points on a wave traveling along

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x direction with $x_2 > x_1$, then the events at x_2 are delayed from the events at x_1 by a time $(x_2 - x_1)/c$ where c is the wave velocity. Thus delay in terms of number of oscillation cycles,

$$n = \frac{(x_2 - x_1)}{cT} \quad \text{where } T \text{ is the time period.}$$

and n is the delay in terms of number of cycles. Thus phase lag (or delay) of x_2 wrt x_1 is simply:

$$\Delta\phi = \frac{x_2 - x_1}{cT} \times 2\pi = \frac{x_2 - x_1}{\left(\frac{\lambda}{T} \times T\right)} \times 2\pi = \frac{2\pi}{\lambda} \Delta x.$$

Thus phase difference :

$$\boxed{\Delta\phi = \frac{2\pi}{\lambda} \Delta x}$$

→ It must be kept in mind that the point requiring longer travel time is lagging in phase and one requiring shorter travel time is leading in phase.

→ If Δt is the time lag between the two points (x_1 and x_2), the phase lag can also be written as:

$$\Delta\phi = \left(\frac{x_2 - x_1}{c}\right) \cdot \frac{2\pi}{T} = \frac{2\pi}{T} \Delta t \quad (\because x_2 - x_1 = c\Delta t)$$

thus $\boxed{\Delta\phi = \frac{2\pi}{T} \Delta t}$

Phase Difference Between Two Waves Reaching A Point Through Differen Routes.

The ideas discussed in previous section can be extended to finding phase difference between two waves reaching same point. If x_1 and x_2 are the paths travelled by two waves ($x_2 > x_1$) then because of the path difference second wave lags behind the first wave by a phase of $\frac{2\pi}{\lambda}(x_2 - x_1)$. If there is some initial lag or lead between the waves then that must be carefully superposed on this lag with proper sign to get the net lag.

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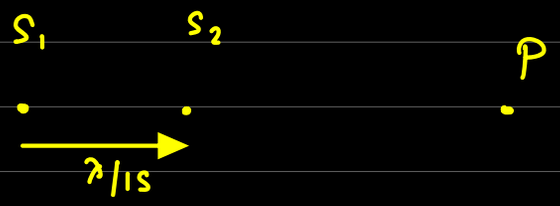
EXAMPLE: Consider two coherent sources S_1 and S_2 with S_1 leading S_2 by a phase of $\pi/2$. Find the phase difference between the waves from S_1 and S_2 when they reach point P.



Solution: For net phase difference we need to superpose initial phase difference upon the phase difference due to path difference. Now

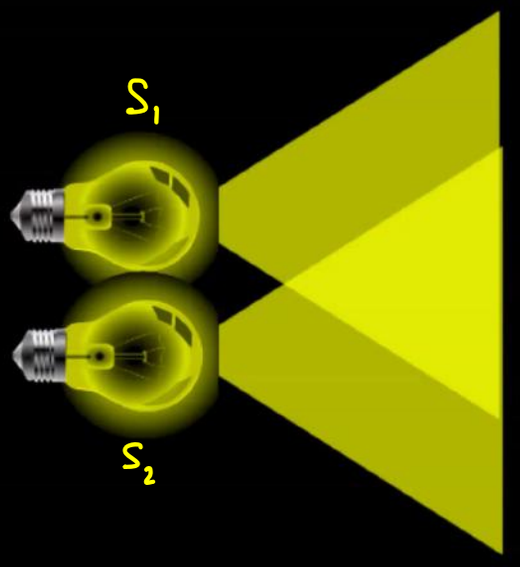
$S_1P - S_2P = \lambda/15 \Rightarrow \Delta\phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{15} = \frac{2\pi}{15}$ with S_1 lagging

But initial lead of S_1 is $\pi/2$ thus net lead of S_1 at P over S_2 remains $\frac{\pi}{2} - \frac{2\pi}{15} = \frac{11\pi}{30}$. Thus at point P, wave from S_1 leads the wave from S_2 by a phase of $\frac{11\pi}{30}$ rad.

→ Note the proper sign while superposing the two phase difference.

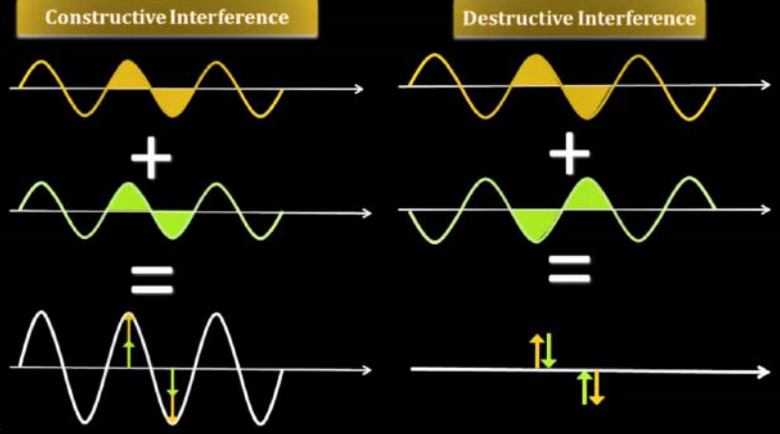
INCOHERENT SOURCES:

Two or more sources are said to be incoherent if they do not maintain a constant phase difference (even if they have same frequency). For incoherent sources intensities get simply added up. i.e if due to two individual sources intensities are I_1 and I_2 at some point then under their simultaneous operation intensity is simply $I_1 + I_2$.



CONSTRUCTIVE & DESTRUCTIVE INTERFERENCE:

If at the point of interference, $\Delta\phi = 2n\pi$ (or $\Delta x = n\lambda$ if all phase difference is due to path difference alone) then we say that interference is constructive and resultant EM oscillations have largest possible amplitudes. While if $\Delta\phi = (2n+1)\pi$



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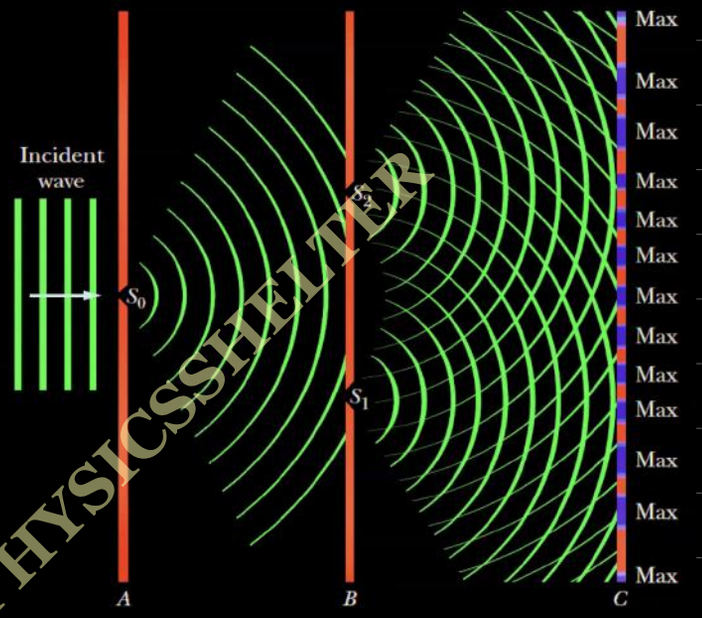
(or $\Delta x = (2n+1)\frac{\lambda}{2}$ if phase difference is due to path difference alone)

then we say that interference is destructive, in this case we get EM oscillations of lowest possible amplitude (or even zero if both waves have same amplitudes).

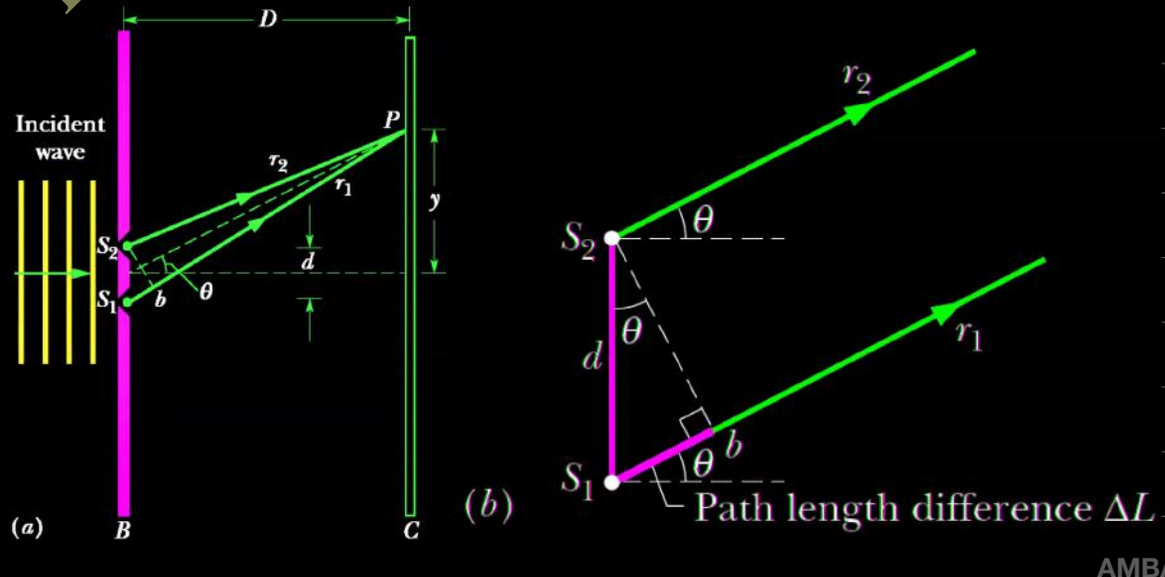
Young's Double Slit Experiment :

In a typical young's double slit experiment, two slits are placed within the diffraction maximum of a single slit. Usually the two slits S_1 and S_2 are placed symmetrically with respect to the diffraction slit S_0 (see adjacent figure). This is a case of interference by division of wavefront. The slits S_1 and S_2 in turn act as sources of secondary wavelets and thus produce an interference pattern on the screen.



→ Since most of the light in slit diffraction is contained in central maximum, it is important that S_1 and S_2 must be in this region otherwise interference pattern on the screen will be very dim or invisible.

Fringe Location, Width & Number Of Fringes In Standard Vdse :



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Consider interference of waves coming from S_1 and S_2 interfering at point P on the screen (figure a above). Typically $D \gg d$ so to find the path difference we can put a compass at P and make an arc of radius PS_2 which cuts S_1P nearly at 90° angle. The enlarged image near this arc is shown in figure (b). From this figure we can say that path difference $\Delta x \cong S_1 b = d\sin\theta$

Further if θ is small enough, $\sin\theta \cong \tan\theta = y/D$ thus with these assumptions we can write:

$$\Delta x = d\sin\theta \cong d\tan\theta = dy/D \quad (1)$$

Now if S_1 and S_2 are equidistant from S_0, there is no initial phase difference, thus for constructive interference, $\Delta x = n\lambda$ thus:

$$\frac{dy}{D} = n\lambda \Rightarrow \boxed{y = \frac{n\lambda D}{d}} \leftarrow \text{location of Maxima.}$$

In this formula, for $n=0$, we call the bright fringe as central maximum while for general $n = \pm i$ we call it i^{th} order maxima, thus for $n = +2$ or $n=-2$ we have 2nd order maxima.

For locating minima we can equate path difference to odd multiple of $\lambda/2$ thus for minima: $\Delta x = (2n-1)\lambda/2$ i.e

$$\frac{dy}{D} = \pm(2n-1)\lambda/2 \Rightarrow \boxed{y = \pm(2n-1)\frac{\lambda D}{2d}} \leftarrow \text{location of minima.}$$

→ In this formula n can be put as 1, 2, 3... and n represents the order of the minima.

Using above formulae, we can list out the maxima locations as:

Maximum: $y = 0, \pm\frac{\lambda D}{d}, \pm\frac{2\lambda D}{d}, \pm\frac{3\lambda D}{d}$ ....

Minima: $y = \pm\frac{\lambda D}{2d}, \pm\frac{3\lambda D}{2d}, \pm\frac{5\lambda D}{2d}$ ....

→ A maximum lies exactly at the mid point of two minima and a minimum lies at the mid point of any two successive maxima.

→ Spacing between two successive maxima or two successive

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minima is called fringe width. Mathematically:

$$\boxed{\beta = \frac{\lambda D}{d}}$$ ← As evident from maxima/minima series above.

Words Of Caution:

→ While solving problems, $\sin\theta \approx \tan\theta$ may not actually be a reasonable assumption. In such cases, Path difference must be carefully calculated by actually taking $\sin\theta$ instead of $\tan\theta$.

→ θ must always be considered as the angle subtended at the mid-point of the two slits. This is especially important when considering the fringes formed in the region directly before the slits. (The logic behind this can be readily verified using Pythagoras theorem and binomial approximation).

Counting The Number Of Fringes:

Since $\Delta x = d\sin\theta$, path difference can have a value between $\pm d$. Thus maximum order of maximum possible is $\text{GIF}(d/\lambda)$. Thus if d/λ is an exact integer, total number of maxima is $\boxed{2\,d/\lambda + 1}$

(counting the one's formed at ∞ and the central maximum). If the maximum at ∞ is to be excluded then number of maxima in such a case will become $2\left[\frac{d}{\lambda} - 1\right] + 1$ i.e

$$\boxed{N = 2\,d/\lambda - 1}$$

→ If d/λ is not an integer then no of maxima is simply:

$$\boxed{N = 2\,\text{GIF}(d/\lambda) + 1}$$

→ Now minima can be found by fitting number of odd multiples of λ within d and doubling the number thus obtained. (Between every two maxima, there is a minimum). The minimum at infinity is usually not counted. If there are N maxima,

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there are either N-1 minima or N+1 minima depending on whether or not a well defined minimum is formed beyond the last maximum or not. (i.e if fractional part of d/λ is more than 0.5 then N+1 minima are formed, otherwise N-1 minima are formed, for exact 0.5, minima exist at ∞ and whether or not they need to be counted must be clearly specified). The following examples will help in understanding the method for counting the number of bright and dark fringes.

→ Suppose $d = 4\lambda$, theoretically 9 maxima exist (with corresponding path differences $0, \pm\lambda, \pm 2\lambda, \pm 3\lambda$ and $\pm 4\lambda$ although the one's corresponding to $\pm 4\lambda$ are formed at ∞ and hence not obtainable on screen. Total minima are 8 corresponding to path differences $\pm 0.5\lambda, \pm 1.5\lambda, \pm 2.5\lambda$ and $\pm 3.5\lambda$. (All obtainable on screen).

→ If $d = 3.75\lambda$, total 7 maxima are formed, all obtainable on screen corresponding to path differences $0, \pm\lambda, \pm 2\lambda$ and $\pm 3\lambda$ and total number of minima is 8 (all obtainable on screen) corresponding to path differences $\pm 0.5\lambda, \pm 1.5\lambda, \pm 2.5\lambda$ and $\pm 3.5\lambda$

→ If $d = 3.5\lambda$, total no of theoretical minima is 8 corresponding to path differences $\pm 0.5\lambda, \pm 1.5\lambda, \pm 2.5\lambda$ and $\pm 3.5\lambda$ however the one's at $\pm 3.5\lambda$ are not obtainable on screen, thus on screen only 6 minima are obtainable. Here no of maxima is 7 corresponding to $x = 0, \pm\lambda, \pm 2\lambda$ and $\pm 3\lambda$ (All obtainable on screen).

→ Since if n is taken as positive, for minima, $d = (2n-1)\lambda/2$ for upper bound of n thus, $n = \left(\frac{2d}{\lambda} + 1\right)\frac{1}{2}$, thus actual value of n is $GIF\left[\frac{d}{\lambda} + \frac{1}{2}\right]$ and thus $\boxed{N_{min} = 2\,GIF\left[\frac{d}{\lambda} + \frac{1}{2}\right]}$

(This includes all theoretically possible minima. If the argument of GIF is exact integer then we need to subtract 2 to get the minima obtainable on screen).

→ For maxima, theoretical number is $\boxed{N_{max} = 2\,GIF\left(\frac{d}{\lambda}\right) + 1}$

if argument of GIF is exact integer then we need to

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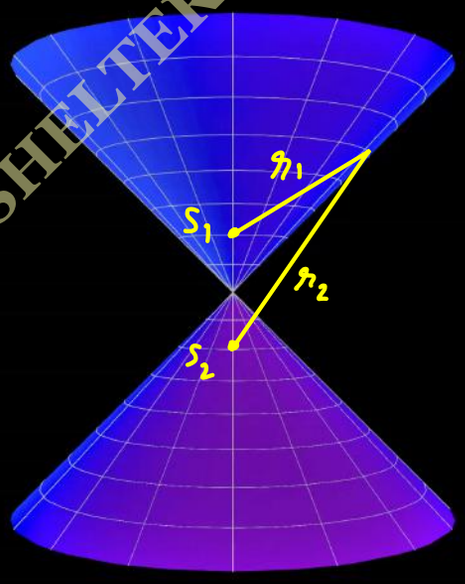
subtract 2 to get maxima obtainable on screen.

ANGULAR POSITION OF FRINGES: Recall that for maxima, $d\sin\theta = n\lambda$ thus angular position of fringes is independent of D. For maxima the angular positions are simply given as $\boxed{\theta_{max} = \pm \sin^{-1}(n\lambda/d)}$

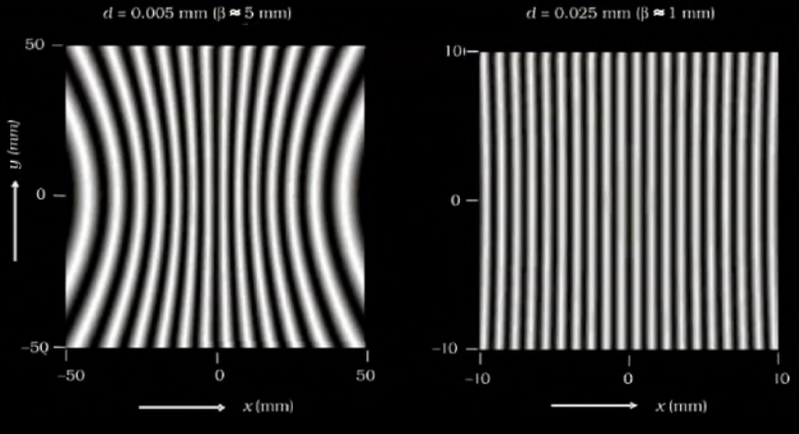
Similarly for minima, $\boxed{\theta_{min} = \pm \sin^{-1}\left[\frac{(2n-1)\lambda}{2d}\right]}$

SHAPE OF FRINGES: The surface $r_2 - r_1 =$ Constant represents the points having same path difference. In 2D this represents a hyperbola, while in 3D this represents a hyperboloid with line joining S_1 and S_2 as the axis of rotation of the hyperbola. Just as we have a pair of straight line serving as asymptotes to hyperbola, in 3D, a conical surface serves as the "asymptotic surface" to the hyperboloid and for large distances, the hyperboloid can be approximated as a double cone with symmetry axis on the line S_1S_2.



Thus for arbitrary screen placement, the fringes will look like conic sections (circles / ellipses / hyperbolae, parabola etc depending on placement of the screen). For standard YDSE set up, the screen is $\perp$ to the axis of the cone and thus the fringes have a hyperbolic shape.

Though if we look only at a small region directly in front of the slits, they may appear like parallel bands but they are portions of hyperbolae with radii of curvature significantly larger than the scale of observation.



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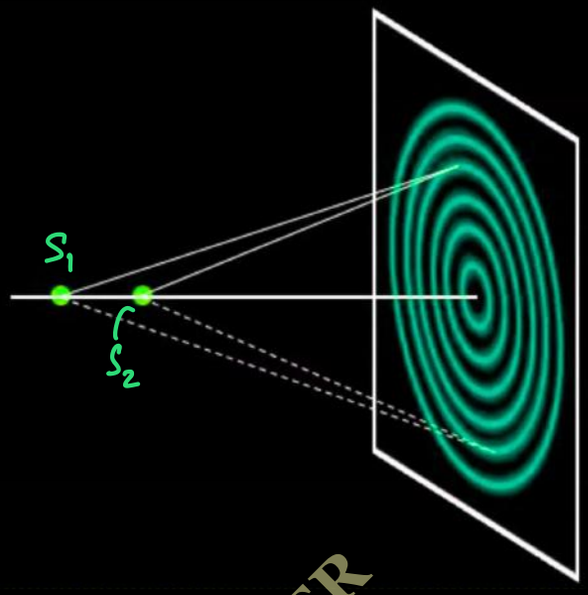
If the screen were instead placed perpendicular to the line joining S_1 and S_2 we will have circular fringes.
(Be careful while counting the fringes here, unlike case of hyperbolic fringes there is no upper and lower first order fringes rather, it is just whole first order circle also, highest order fringes have smallest radii because largest path difference occurs near centre of the screen).



INTENSITY VARIATION IN YDSE:

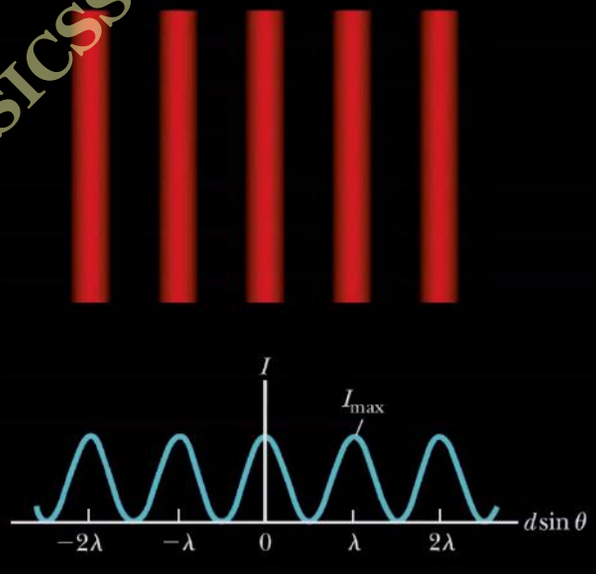
Let I_0 be the individual intensity at any point due to each individual slit then, due to combined effect of both slits,

$I = I_0 + I_0 + 2I_0 \cos\phi$

$= 2I_0 [1 + \cos\phi]$

$= 4I_0 \cos^2(\phi/2)$, Thus:

$$\boxed{I = 4I_0 \cos^2(\phi/2)}$$



Now $\Delta\phi = \frac{2\pi}{\lambda} \cdot \Delta x = \frac{2\pi}{\lambda} \cdot d\sin\theta \approx \frac{2\pi}{\lambda} \cdot \frac{dy}{D}$ but $\frac{\lambda D}{d} = \beta$

thus $\Delta\phi = 2\pi y/\beta$ thus: $\boxed{I = 4I_0 \cos^2 \pi y/\beta}$

→ This formula is often more handy when y is known as a multiple of β i.e fringe width.

→ This formula uses the approximation $\sin\theta \approx \tan\theta$ so must be used with caution if θ is large.

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Example: In YDSE, two slits are 1mm apart and illuminated by a light of wavelength 500 nm. Find (a) Fringe width (b) location of 3rd bright fringe (c) distance between 4th and 6th bright fringe on same side of central maximum (d) Ratio of intensities at points $y_1 = \beta/4$ and $y_2 = \beta/3$. The screen is 2m away from slit plane.

Solution: $d = 1mm$, $\lambda = 500nm$, $D = 2m$ (a) $\beta = \frac{\lambda D}{d} = \frac{5\times10^{-7}\times2}{10^{-3}} = 1mm$ Ans.

(b) Third bright fringe is located at 3β i.e $y_3 = 3mm$ Ans.

(c) Distance between 4th and 6th bright fringe is 2β i.e 2mm.

(d) $I = 4I_0\cos^2(\pi y/\beta)$, thus, $I_1/I_2 = \cos^2\left(\pi\times\frac{\beta/4}{\beta}\right)/\cos^2\left(\pi\times\frac{\beta/3}{\beta}\right)$

i.e $I_1/I_2 = \cos^2(\pi/4)/\cos^2(\pi/3) = \frac{1}{2}\times4 = 2$ Ans.

Effect Of Immersing The YDSE Setup In Liquid:

If λ is wavelength of light in air, then in a liquid of refractive index μ, the wavelength decreases by a factor of μ i.e $\lambda_l = \lambda/\mu$ where λ_l represents wavelength in the liquid. When we immerse the YDSE setup in a liquid, all the calculations and methods of derivation remain the same except that everywhere we need to replace λ by λ_l.

Example: A YDSE is performed inside water ($\mu = 4/3$) with light of frequency 6×10^{14} Hz. If the slits are separated by 0.2 mm and screen distance is 1m, find the fringe width and total number of bright and dark fringes.

Solution: $\beta = \lambda_l D/d \Rightarrow \frac{c}{\mu\nu}\frac{D}{d} = \frac{3\times10^8}{\frac{4}{3}\times6\times10^{14}}\times\frac{1}{2\times10^{-4}} = 1.88mm.$

Now $\lambda_l = \frac{c}{\mu\nu} = \frac{3\times10^8}{\frac{4}{3}\times6\times10^{14}} = 375\,nm.$

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Now $N_{max} = 2\,GIF(d/\lambda_\ell)+1 = 2\,GIF(2\times10^{-4}/375\times10^{-9}) + 1$

$= 2\,GIF(533.33)+1 = 1067$

Thus number of bright fringes in water: 1067 nos.

Now no of minima will correspond to the series $0.5, 1.5 \cdots 532.5$ i.e 533 above and 533 below central maximum i.e 1066 thus number of minima is 1066 nos. Alternatively we can calculate minima using:

$$N_{min} = 2\,GIF\left(\frac{d}{\lambda} + \frac{1}{2}\right)$$

thus $N_{min} = 2\,GIF\left[533.33 + 0.5\right] = 2\times533$ i.e 1066 nos.

→ Think how many maxima and minima will be obtained for same set up in air.

Solution: $\lambda = \lambda_\ell \times \mu = 375nm \times 4/3 = 500nm$. Thus,

$N_{max} = 2\,GIF\left(d/\lambda\right) + 1 = 2\,GIF\left(2\times10^{-4}/500\times10^{-9}\right) + 1$

$= 2\,GIF(400) + 1 = 801$ nos.

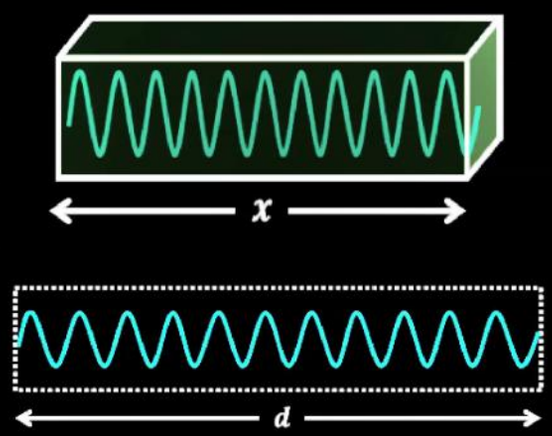
However ∵ argument of GIF is exact number, the actual maxima obtainable on screen is 801 - 2 i.e 799 nos.

Now $N_{min} = 2\,GIF\left(d/\lambda + \frac{1}{2}\right) = 2\,GIF(400.5) = 800$ nos.

[Alternatively, minima belong to the series $0.5, 1.5 \cdots 399.5$ thus there are 400 minima above and 400 minima below the central max.

OPTICAL PATH: After traveling a distance x in a medium of refractive index μ, a wave lags by the same number of cycles as it would have lagged in travelling a distance μx in vacuum. This quantity μx is called the optical path (or vacuum equivalent path). Thus:

$\boxed{d = \mu x}$ ← Optical Path.



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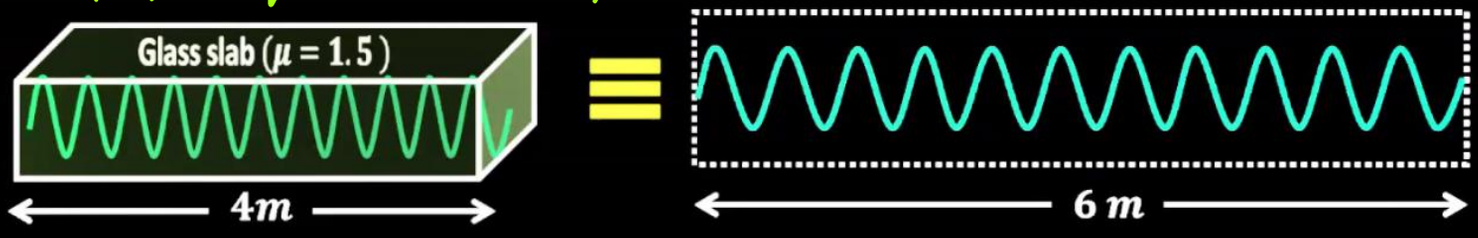
→ As an example, a path length of 4m travelled in a glass slab is equivalent to travelling 6m (i.e 4x1.5) in vacuum for purpose of calculating the phase las.



→ For a wave travelling in vacuum, optical path length is equal to the geometrical path length.

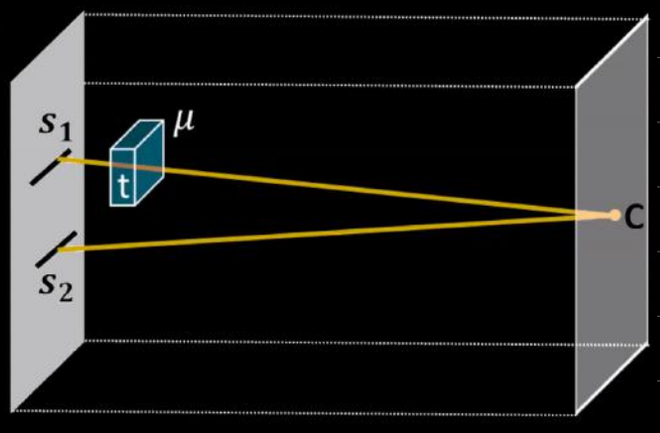
→ Optical path lengths of all paths between an object and its image are equal (Fermat's principle ... recall degenerate minima idea).

MEANING OF "CENTRAL MAXIMUM": In many problems involving modifications of YDSE we come across the term "central Maximum"; this always means the location of the fringe where phase difference between interfering waves is zero and if the phase difference is happening only because of optical path length, then central maximum is located where the optical path difference between interfering waves is zero.

EFFECT OF PUTTING A THIN FILM IN FRONT OF THE SLITS:

Putting a thin film in front of one of the slits for YDSE set up in air shifts the central maximum towards the slit where film is inserted.

Now by definition of central maximum, $S_1C_{optical} = S_2C_{optical}$ thus:



$$[S_1C - t + t\mu] = S_2C \Rightarrow S_2C - S_1C = (\mu - 1)t = d\sin\theta$$

where θ is the angular position of central maximum thus:

$$\boxed{\sin\theta = (\mu-1)\frac{t}{d}}$$

this gives the angular position of new central maximum. Now if θ is small, $\sin\theta \approx \tan\theta$

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$= y_c/D$ then, $\frac{y_c}{D} = (\mu-1)\frac{t}{d} \Rightarrow \boxed{y_c = (\mu-1)\,t\,\frac{D}{d}}$

The fringe width still remains same.

→ While calculating the effect of thin films, we ignore the bending of light upon refraction through the film, which is a reasonable assumption so long as θ is small.

→ There is no effect on the fringe width. Only the fringe pattern is offset by the distance y_c.

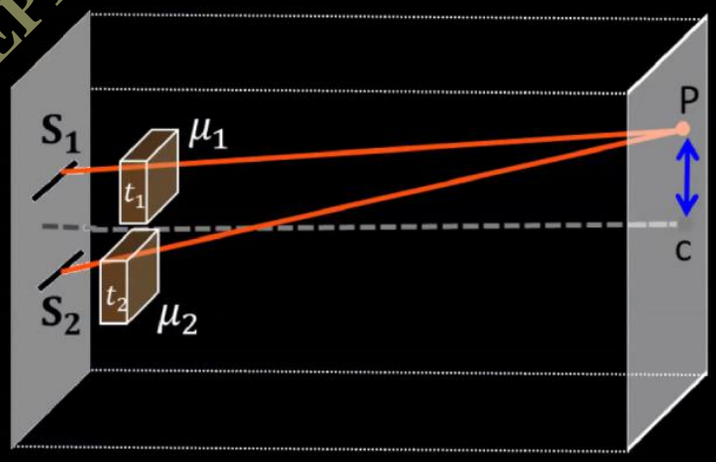
→ Sometimes we are required to calculate the number of fringes by which the pattern shifts, this is simply obtained by dividing the fringe shift by the fringe width i.e $N = y_c/\beta = ((\mu-1)\,t\,D/d)/(\lambda D/d)$ i.e

$$\boxed{N = (\mu-1)\,t/\lambda}$$

PLACING FILMS BEFORE BOTH SLITS:

If a film is placed before both slits, net effect can be found by superposition, using the logic presented above. Due to upper slit, number of upward fringe shift:

$N_1 = (\mu_1-1)\,t_1/\lambda$ similarly due to lower film, number of downward fringe shift $= (\mu_2-1)\,t_2/\lambda$. Thus if



$(\mu_1-1)\,t_1 > (\mu_2-1)\,t_2$ there is net upward shift, for no net shift,

$(\mu_1-1)\,t_1 = (\mu_2-1)\,t_2$. (i.e increase in optical path to the mid point of screen is same for both waves).

EXAMPLE: In YDSE, $\lambda = 400$ nm is used when a film of $\mu = 1.5$ and $t = 6\,\mu m$ is placed in front of the upper slit. (a) Find the number of maxima that crossed the centre line. (b) What should be thickness of a film having refractive index 2 that should be

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placed before the lower slit to restore the position of central maximum?

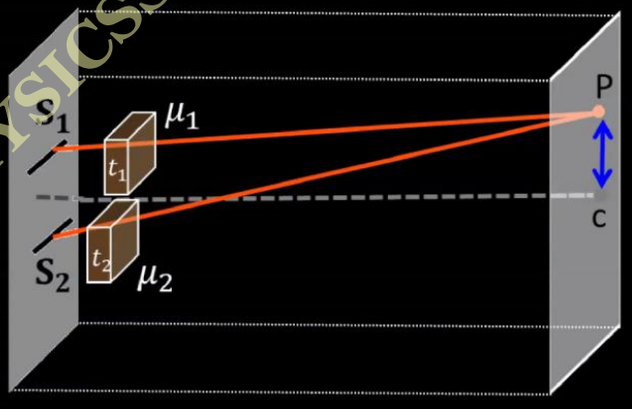
Solution: $N = (\mu-1)\,t/\lambda = (3/2-1) \times \dfrac{4\times10^{-6}}{4\times10^{-9}} = 5$

Thus the fringe pattern shifts by 5β, i.e the 5^{th} order fringe that was below the central maximum, now occupies the position of central maximum. (5 fringes have crossed).

Now for restoring the location of central maximum, $(\mu_1-1)\,t_1 = (\mu_2-1)\,t_2$

Thus $0.5 \times 4\,\mu m = 1 \times t_2 \Rightarrow t_2 = 2\,\mu m$ Ans.

EXAMPLE: Suppose thin films were put in front of slits as shown and then entire apparatus were immersed in a liquid of refractive index μ, what would be the angular position of central maximum.



Solution: METHOD I : OPTICAL PATH:

For central maximum: $(S_1P - t_1)\mu + \mu_1 t_1 = (S_2P - t_2)\mu + \mu_2 t_2$

$\Rightarrow \mu\,(S_2P - S_1P) = \mu\,(t_2 - t_1) + \mu_1 t_1 - \mu_2 t_2$

$\Rightarrow \mu\, d\sin\theta = \mu\,(t_2 - t_1) + \mu_1 t_1 - \mu_2 t_2$

$\Rightarrow \sin\theta = \dfrac{1}{d}\left[\, t_2 - t_1 + \dfrac{\mu_1 t_1 - \mu_2 t_2}{\mu}\right]$ Ans.

i.e $\sin\theta = \dfrac{1}{d}\left[\,(\mu_{1r}-1)\,t_1 - (\mu_{2r}-1)\,t_2\right]$ where $\mu_{1r} = \mu_1/\mu$, $\mu_{2r} = \dfrac{\mu_2}{\mu}$

(Note that even if there were no fringe shift with apparatus in air, there can be fringe shift when immersed in liquid since here for no fringe shift, $(\mu_{1r}-1)\,t_1 = (\mu_{2r}-1)\,t_2$

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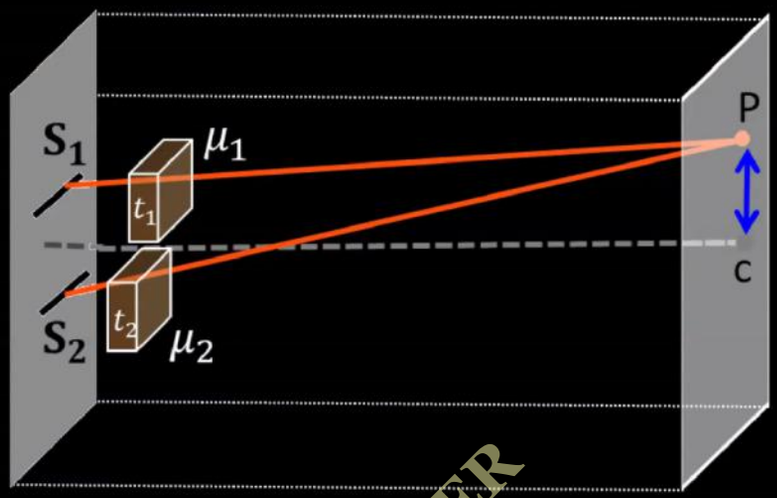
Method 2: Zero Time Difference:

For zero phase difference, the waves from S_1 and S_2 must reach at the same time, thus:

$$\frac{S_1P - t_1}{v} + \frac{t_1}{v_1} = \frac{S_2P - t_2}{v} + \frac{t_2}{v_2}$$



where v, v_1 and v_2 represent speed of light in media μ, μ_1 and μ_2.

$$\text{Thus } \frac{(S_2P - S_1P)}{v} = \frac{t_2 - t_1}{v} + \frac{t_1}{v_1} - \frac{t_2}{v_2}$$

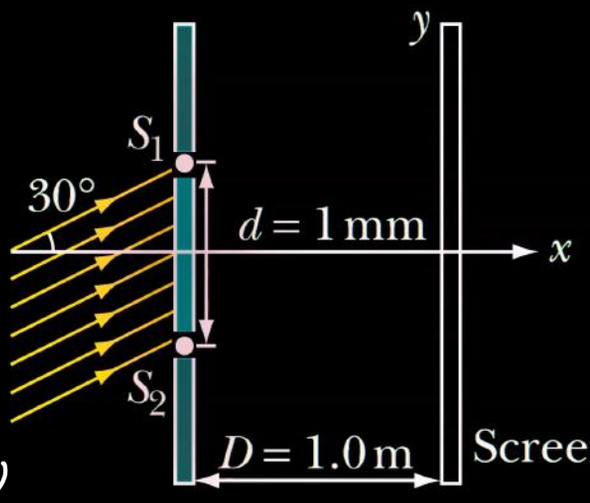
Now $S_2P - S_1P = d\sin\theta$, $v = c/\mu$, $v_1 = c/\mu_1$ and $v_2 = c/\mu_2$ thus:

$$\mu(d\sin\theta) = \mu(t_2 - t_1) + \mu_1 t_1 - \mu_2 t_2 \quad \text{or,}$$

$$\sin\theta = \frac{1}{d}\left[(\mu_{12} - 1)t_1 - (\mu_{21} - 1)t_2\right] \quad (*\text{Same result as above}).$$

YDSE With Asymmetric Source Location:

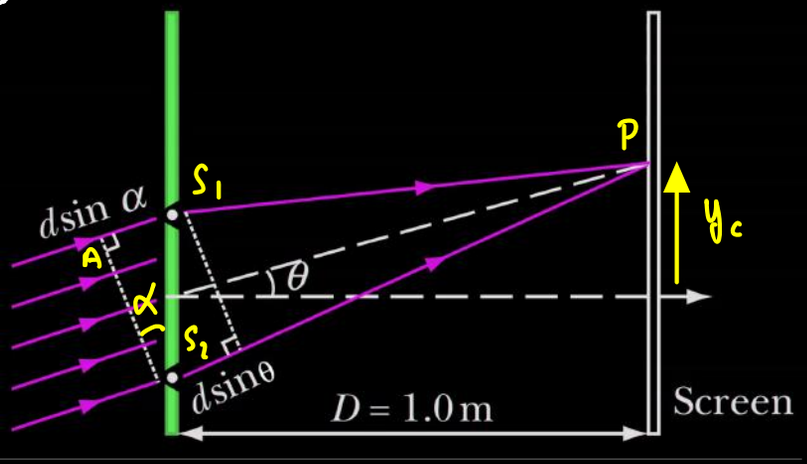
A coherent beam of wavelength $\lambda = 0.5$ mm falls on a YDSE set up. Slit separation is 1mm and screen distance is 1m. If the incident beam makes an angle 30° with x-axis as shown, find the coordinate of first minima on either side of central maximum. (Also locate central max)



Solution: We know that wavefronts are surfaces of same phase. Thus imagining a wavefront AS_2, the path lengths from this wavefront to central maximum must be equal thus $AS_1P = S_2P$ thus:

$AS_1 + S_1P = S_2P \Rightarrow AS_1 = S_2P - S_1P$

Thus $d\sin\alpha = d\sin\theta$ or $\theta = \alpha$.



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Thus with inclined beam, central maximum lies along the incident beam direction from the mid point of the slits. ∴ central maximum shifts by $D\tan\theta = D\tan\alpha = 1 \times \tan 30° = \frac{1}{\sqrt{3}}$ m ($\because \alpha = 30°, D = 1$m).

→ Now be cautious, 30° is not a small angle so we can't just take $\beta = \lambda D/d$. (That is based on $\sin\theta \approx \tan\theta$). Now for first order minimum below central maximum, $AS_1 P_1 - S_2 P = \lambda/2$ thus:

$$d\sin\alpha - d\sin\theta = \lambda/2 \Rightarrow d\sin\theta = d\sin\alpha - \frac{\lambda}{2} \Rightarrow \sin\theta = \sin\alpha - \frac{\lambda}{2d}$$

thus $\sin\theta = \sin 30° - \frac{0.5}{2} = \frac{1}{4} \Rightarrow \tan\theta = \frac{1}{\sqrt{15}}$ thus

the first order minimum below the central maximum is located at $D\tan\theta$ i.e $\frac{1}{\sqrt{15}}$ m from mid point of the screen abv.

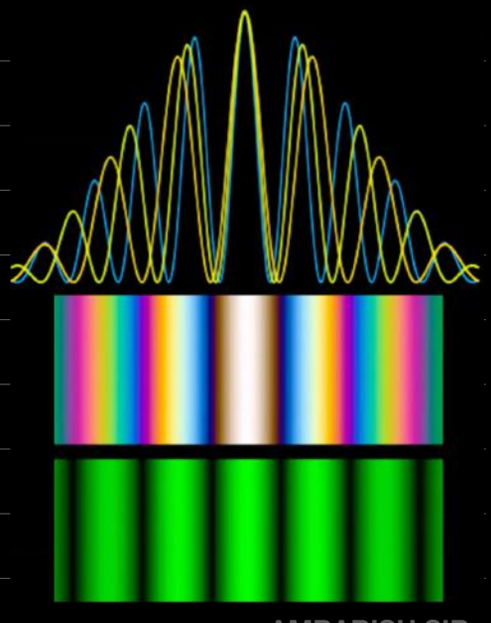
Now for first order minimum above central maximum,

$$S_2P - AS_1P = \lambda/2 \Rightarrow S_2P - (AS_1 + S_1P) = \lambda/2$$

$$\Rightarrow d\sin\theta - d\sin\alpha = \lambda/2 \Rightarrow \sin\theta = \sin\alpha + \frac{\lambda}{2d} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$\Rightarrow \tan\theta = 3/\sqrt{7}$ thus first order minimum above central maximum is located at $D\tan\theta$ i.e $3/\sqrt{7}$ m above mid point of the screen abv.

YDSE WITH WHITE LIGHT: If we perform YDSE with white light, the central fringe is white because all colors simultaneously form a maxima there. Now since blue color has shortest wavelength, its intensity quickly declines as one moves out from the central maximum, while the red color is slowest in its decline due to largest fringe width, accordingly, as one moves out of the centre, one sees a brownish hue around the central maximum (superposition of Red to Green wavelengths in various proportions). After



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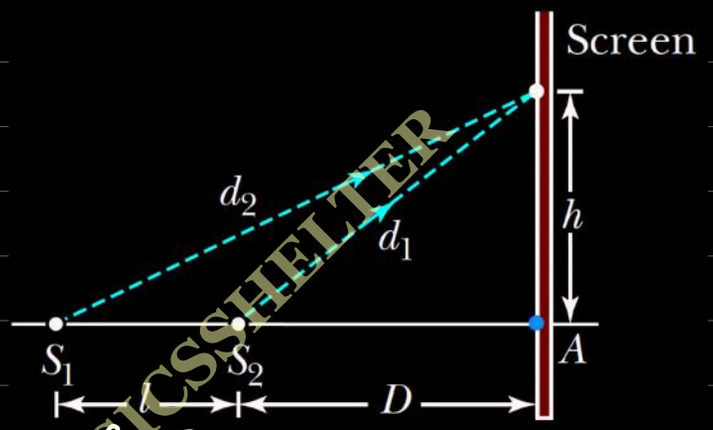
that once again blue attains its maximum and we see a vibgyor pattern which gets gradually washed out as colors begin to mix due to varying fringe widths and far away from centre, once again we see whitish illumination due to mixing of various orders of maxima of different wavelengths spread out over the spectrum.

EXAMPLE: Two point sources oscillating in same phase are arranged on a line perpendicular to a screen ($D >> \lambda$). (a) What is the shape of interference fringes (b) What is the radius of smallest bright fringe (not counting the centre) (c) How many bright and dark fringes are formed (d) What is the radius of first order maximum. Given $l << D$.



$k = n\lambda$

$n >> 1$

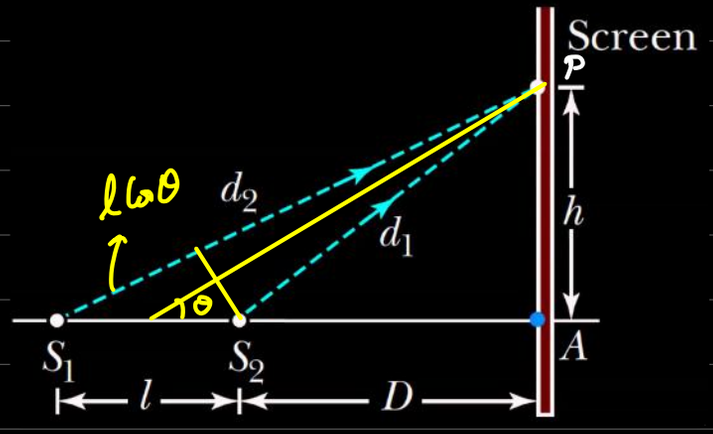
Solution: Here highest order maximum is formed at the centre because here the path difference is maximum and it goes to zero as one moves outward to infinity. Thus as one moves out from centre of the screen, the fringe order decreases. The fringes are circular in shape (Think about the double cone axis).

(b) Path difference at the centre is $n\lambda$, so there is brightness at the centre which will gradually darken as one moves out and again brighten as one meets $(n-1)$ order maximum. This is the maximum we are looking for.

Thus for $(n-1)$ order maximum:

$$\sqrt{(n\lambda + D)^2 + h^2} - \sqrt{D^2 + h^2} = (n-1)\lambda$$

we can solve this equation for h directly and apply appropriate approximations. Alternatively we could find path difference approximately by dropping a $\perp$ from S_2 P to



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S, P at point S_2 as shown.

Now $l\cos\theta = (n-1)\lambda \Rightarrow \cos\theta = \frac{(n-1)\lambda}{l} = \left(\frac{n-1}{n}\right)$ $(\because l = n\lambda)$.

$$\Rightarrow \tan\theta = \frac{\sqrt{n^2-(n-1)^2}}{n-1} = \frac{\sqrt{2n-1}}{(n-1)} = \sqrt{\frac{2n-1}{n^2-2n+1}} \approx \sqrt{\frac{2}{n}}$$

Assuming $n >> 1$. Now $h \approx D\tan\theta \approx (D+l)\tan\theta \approx \sqrt{D(D+l)}\tan\theta$

thus:

$$h = \sqrt{\frac{2D(D+n\lambda)}{n}} = \sqrt{2D\lambda\left(\frac{D}{l}+1\right)} \text{ Ans.}$$

(c) Now min path difference can be zero and maximum $n\lambda$ thus theoretical number of possible bright rings is the number of terms in the sequence $0, 1, 2 \ldots n$ i.e $(n+1)$ thus total number of theoretical bright rings is $(n+1)$, though one ring is formed at ∞, ignoring that we have n bright rings (including the one point ring of zero radius at centre of the screen).

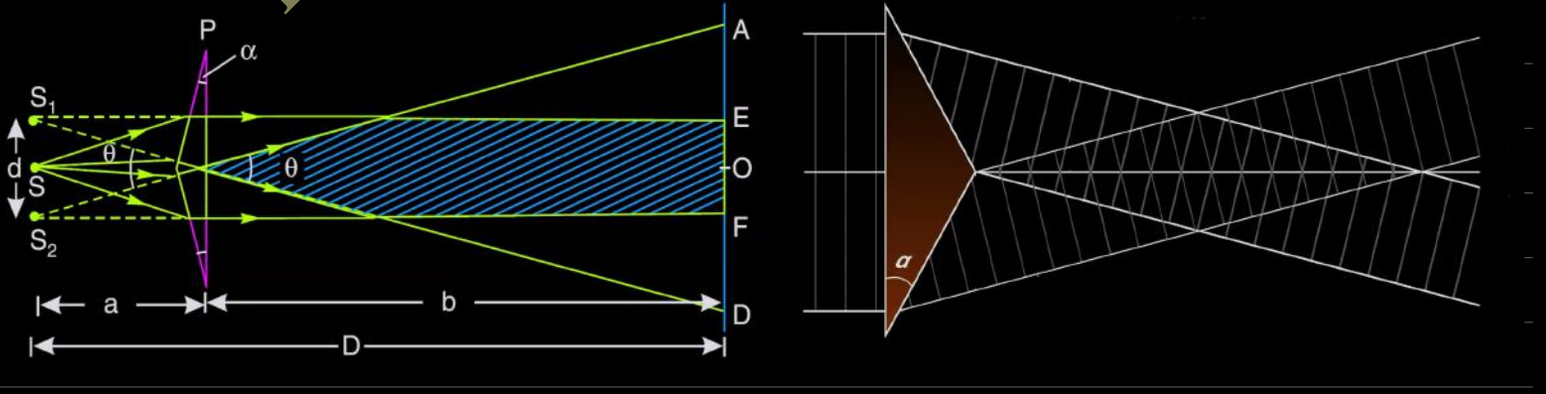
For dark fringes, we need to count no of terms in series $0.5, 1.5, \ldots (n-0.5)$ i.e n, thus total n dark fringes are formed.

(b) For first order maximum, $l\cos\theta = \lambda \Rightarrow \cos\theta = \frac{1}{n} \Rightarrow \tan\theta = \frac{\sqrt{n^2-1}}{\approx n}$

Thus required ring radius: $h \approx D\tan\theta \approx (D+l)\tan\theta$

$\approx \sqrt{D(D+l)}\tan\theta = n\sqrt{D(D+n\lambda)}$ Ans.

FRESNEL BIPRISM ARRANGEMENT:



In this arrangement, two images of a source are formed using two thin prisms and then the two images act as coherent sources just like slits S_1 and S_2 in YDSE.

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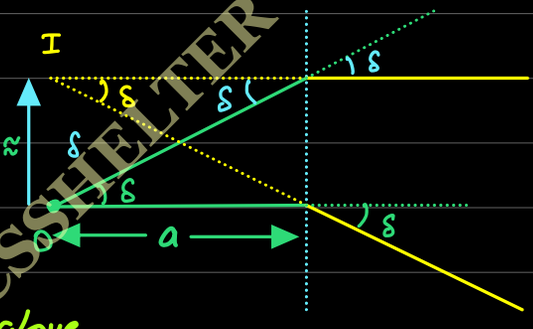
→ The interference pattern is obtained only in the region where light from both the prisms reaches.

→ Depending on source location and height of the prisms, the region where interference is possible could either be increasing, constant or first increasing and then decreasing as the screen is moved away from the prisms. (For instance in figure 2, a kite shaped region is obtained where interference happens). If the screen is placed beyond this region, no fringes are obtained.

→ Recall image formation from a thin prism. The image is formed vertically at a distance $a\tan\delta \approx a\delta$ i.e $a(\mu-1)A$ above the source thus,

$$\boxed{\alpha = 2a(\mu-1)A}$$ ← See first figure above



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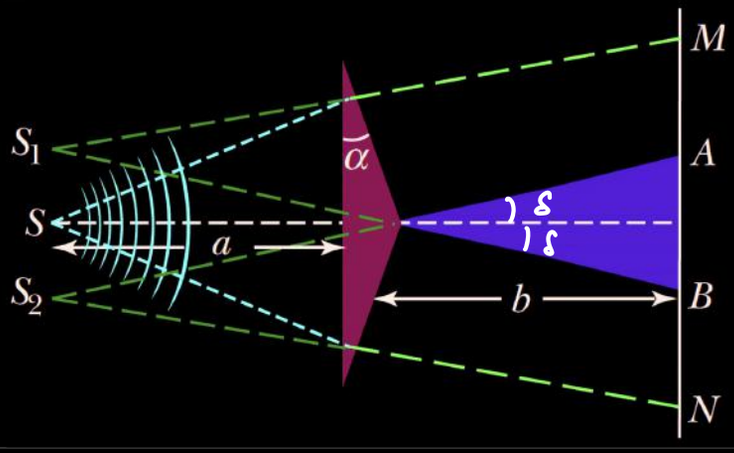
EXAMPLE: In the fresnel biprism arrangement shown, $a = 1m$, $b = 4m$, $\alpha = 2\times10^{-3}$ rad, $\lambda = 6000\,\text{Å}$ and $\mu = 3/2$. (a) Find the fringe width (b) How many fringes can be observed on the screen if the prisms have sufficient height.



Solution: (a) $\alpha = 2a(\mu-1)A = 2\times1\times\frac{1}{2}\times2\times10^{-3} = 2\,\text{mm}$. $D = 1+4 = 5$

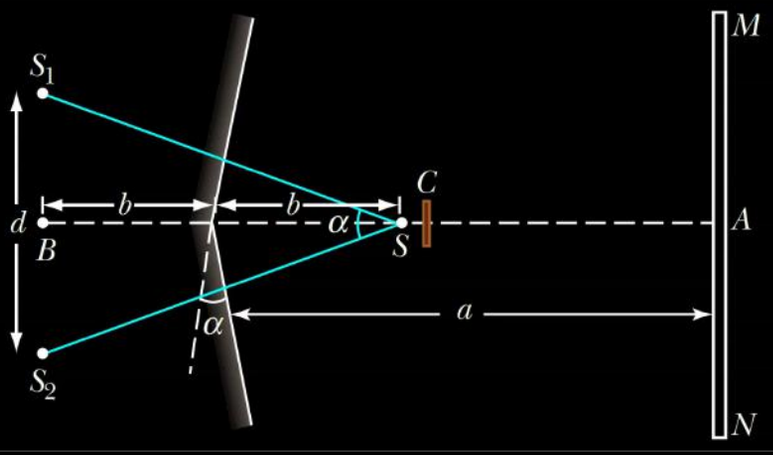
$$\Rightarrow \beta = \frac{\lambda D}{\alpha} = \frac{6\times10^{-7}\times5}{2\times10^{-3}} = 15\times10^{-4} = 1.5\,\text{mm}$$ Ans.

(b) Fringes can be formed only between lowest ray from top prism and highest ray from the lower prism. Thus available region for interference $AB = 2bS = 2b(\mu-1)A = 2\times4\times\frac{1}{2}\times2\times10^{-3} = 8\,\text{mm}$.

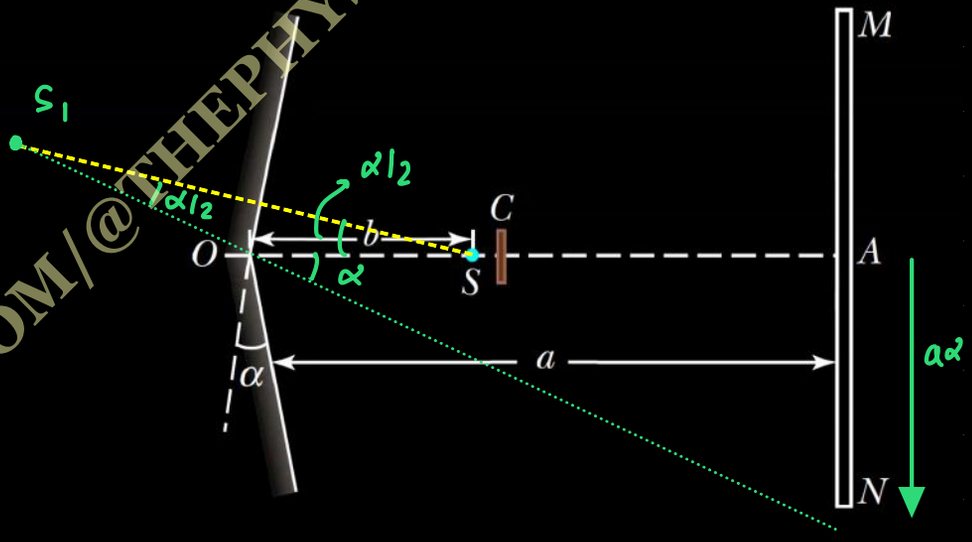
Thus number of fringes $N = \left[\frac{AB}{\beta}\right] = \left[\frac{8}{1.5}\right] = 5$ Ans.

FRESNEL TWO MIRROR ARRANGEMENT:

In this arrangement, two mirrors are kept at a small angle α from each other. The images formed in the two mirrors serve as the coherent sources (S_1 and S_2) (see adjacent figure). If b is the distance of the source from the common edge of the mirrors, then equivalent $\boxed{d = 2b\alpha}$ ($\because \alpha$ is very small). The interference on the screen is observed in the region which recieves reflected light from both mirrors (i.e between lowest ray from upper mirror and the highest ray from the lower mirror).



→ From the adjacent figure, it is clear that reflected ray hits at a distance $a\alpha$ below the mid point of the screen thus the region of the screen where fringes are formed has a width $2a\alpha$.



→ Fringe width $= \lambda D/d = \dfrac{\lambda(a+b)}{2b\alpha}$, thus $\boxed{\beta = \dfrac{\lambda(a+b)}{2b\alpha}}$

→ Now number of fringes $\simeq N = \dfrac{2a\alpha}{\beta} = \dfrac{4ab\alpha^2}{\lambda(a+b)}$

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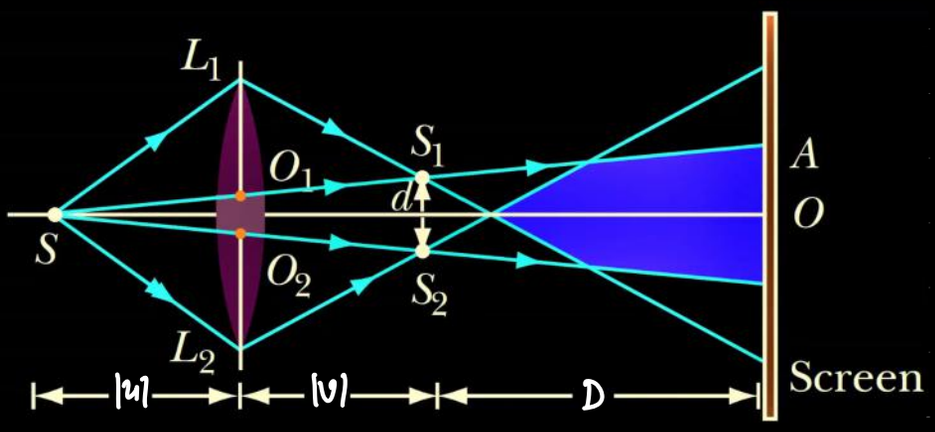
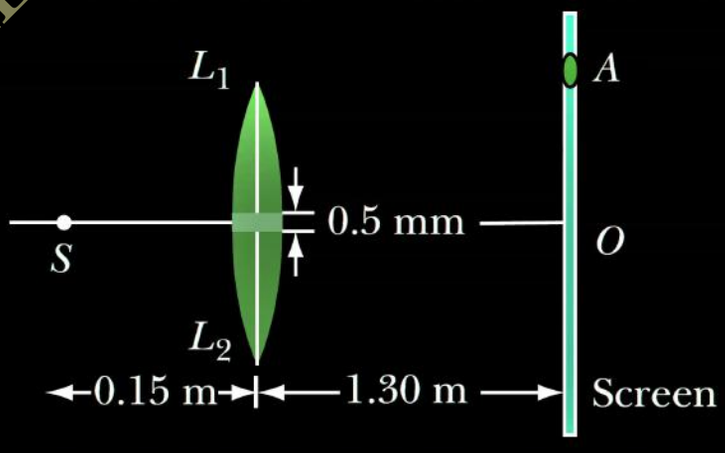
BILLET'S SPLIT LENS ARRANGEMENT :

In this arrangement, a lens is divided in to two halves and the two halves are separated to have parallel principal axes. The two lenses form two images of the source S. S_1 and S_2 thus formed serve as coherent sources for interference on the screen. The interference is observed only in the region recieving refracted light from both the halves (see figure). The distance between S_1 and S_2 (d) and D can be worked out using lens formulae that you learnt in geometrical optics.



EXAMPLE: A lens of focal length 10 cm is cut in to two halves and the two halves are separated by 0.5 mm as shown. A source of $\lambda = 500$ nm is placed 15 cm away from lens, while screen is 130 cm away as shown. Find (a) fringe width (b) OA where A is point of third maximum (c) Width of the region where interference fringes are formed.

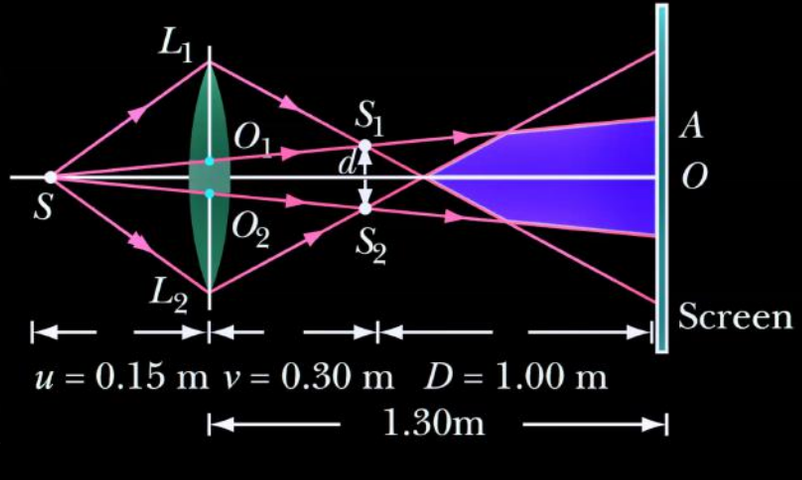


Solution: Using lens formula,

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-15} = \frac{1}{10}$$

$\Rightarrow v = 30$ cm. Now $m = v/u = -2$

Thus, since source is 0.25 mm from each principal axis, its image is 0.5 mm away on other side



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of principal axes. Thus separation between S_1 & S_2 : $(0.25 + 0.5) \times 2$ i.e 1.5 mm. Thus $d = 1.5$ mm, $D = 130\text{ cm} - 30\text{ cm} = 100$ cm.

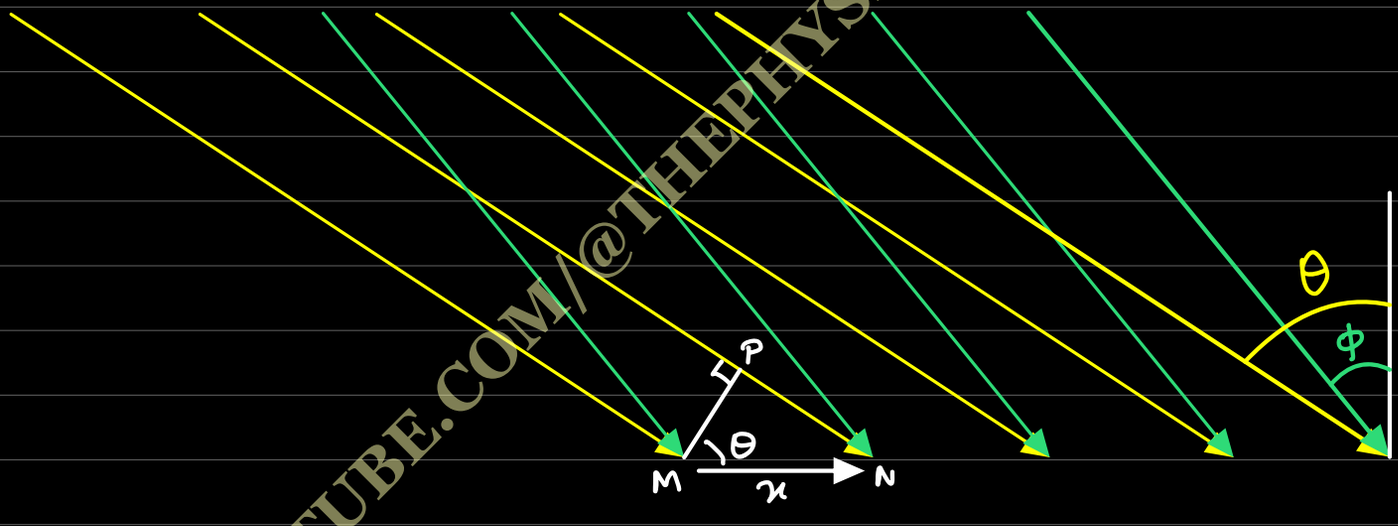
Now $\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{1.5 \times 10^{-3}} = 3.33 \times 10^{-4}$ m. Thus location of 3rd max:

$OA = 3\beta = 1$mm Abv. Now angle subtended by interference region on source is same as that subtended by $S_1 S_2$ i.e $\alpha = \frac{1.5\text{mm}}{450\text{ mm}}$

Thus width of interference region $w = DO \times \alpha = 145\text{cm} \times \frac{1.5}{450}$

$= 4.83$ mm Abv. (No of fringes: $\left\lfloor \frac{4.83\text{mm}}{0.33\text{ mm}} \right\rfloor = \lfloor 14.5 \rfloor = 14$ fringes.

Interference Between Coherent Parallel Beams:



Consider two coherent parallel beams falling on a screen at angles θ and ϕ to the normal of the screen. Let M be a point of maximum and we wish to find the location of the next maximum. Between successive maxima, path difference must change by λ. Now consider a point N at a distance x from M. Here the path of the yellow beam is longer by amount PN (see the wavefront MP). i.e the path of yellow beam has increased by $x \sin\theta$, similarly the path of the green beam has increased by an amount $x \sin\phi$ (Note that wavelengths of two beams are same, colors are chosen only for clear illustration). Thus net increase in path difference, $\Delta l = x \sin\theta - x \sin\phi$

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Now for N to be the next maximum this must be equal to λ thus $x[\sin\theta - \sin\beta] = \lambda \Rightarrow x = \dfrac{\lambda}{\sin\theta - \sin\phi}$

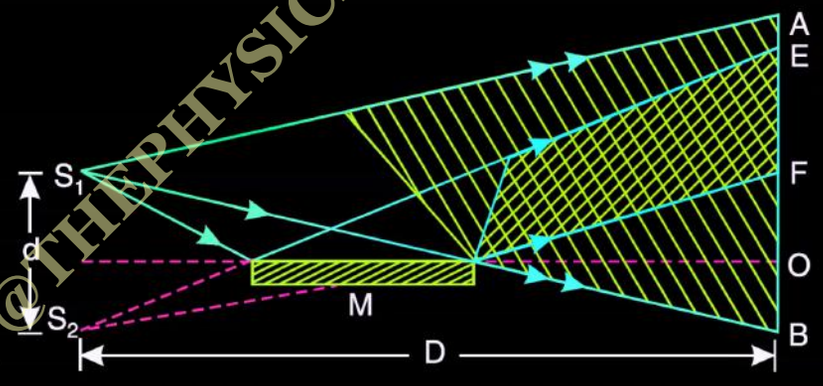
Thus fringe width: $\boxed{\beta = \dfrac{\lambda}{\sin\theta - \sin\phi}}$

→ Here we had chosen θ and ϕ on same side of normal if they were on opposite sides, the denominator will be $\sin\theta + \sin\phi$.

→ If θ and ϕ are small, such that $\theta - \phi = \delta$ then $\beta = \lambda/\delta$, (δ is the angle of divergence between the beams).

Lloyd's Single Mirror Setup:

In this setup a source S_1 is reflected in a mirror to form its image S_2 and then S_1 and S_2 serve as coherent sources to produce interference on screen. Interference pattern is observed only in the region where direct light and reflected light, both are present (See region EF in the figure). This is same as the region of field of view of the image S_2 on the screen. The distance S_1S_2 serves as d and now fringe pattern is calculated just like we do it for YDSE. There are some remarkable distinctions between YDSE and Lloyd's mirror set up as follows:



→ Upon reflection, EM waves undergo π shift thus conditions and locations of maxima and minima get swapped when compared to YDSE.

→ There are no interference fringes below the mirror considering the mirror to be infinite all the way up to screen thus number of fringes must be counted very carefully (in general they will be different from the number in YDSE).

→ Field of view has to be calculated for counting fringes.

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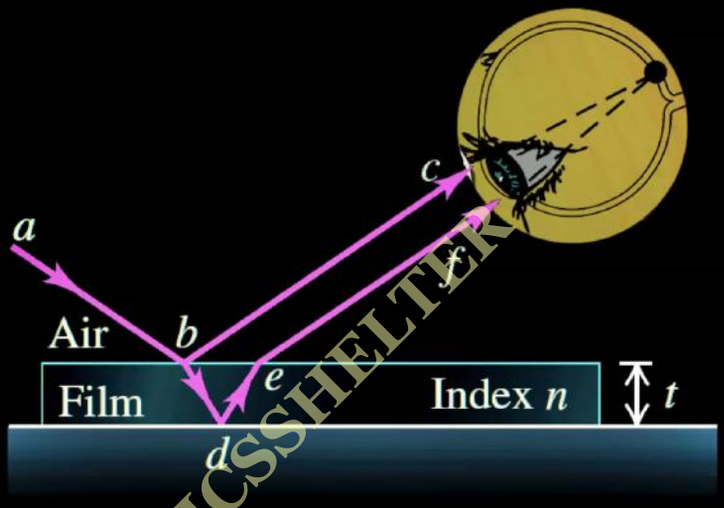
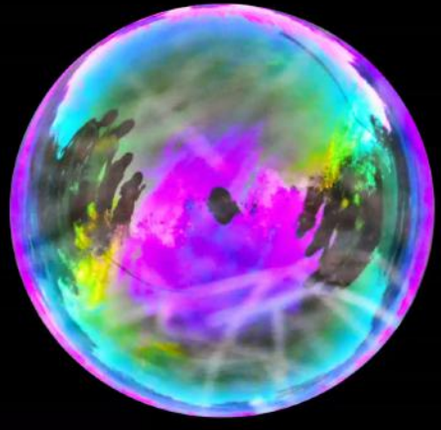
→ The region in front of mid point of S_1 and S_2 is dark ($\because \Delta x = 0$ so only π shift matters). [Assuming full mirror all the way up to O].

THIN FILM INTERFERENCE (DIVISION OF AMPLITUDE):



Consider a ray of light ab falling on a thin film (say of oil). A portion of light is reflected from the top surface while another portion is refracted and a portion of which is reflected again at second interface to refract again producing ray ef. (Actually the process undergoes indefinitely but rays become less and less intense on proceeding). Now the parallel rays bc and ef enter the eyes of the observer to produce interference at the retina. The phase difference between the rays depends on optical paths taken by rays and also on the phase changes due to reflections (Reflection from a denser medium causes π shift while from rarer medium produces no π shift). If the incident light is white, different wavelength will form a maximum at different viewing angles, thus producing the VIBGYOR spectrum. [For instance during rainy season, sometimes a drop of petrol spreads on water, thus forming a thin film, this film shows the rainbow colors depending on the viewing angles.

→ This phenomenon is also responsible for rainbow colors appearing on soap bubbles (Interference of light reflected from top and bottom surfaces of the film).



→ For normal incidence, amplitude of reflected light is related to that of incident light as: $E_r = [(\mu_i - \mu_r)/(\mu_i + \mu_r)] E_i$ where μ_i and μ_r

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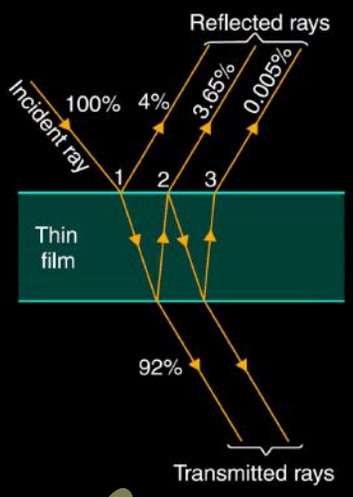
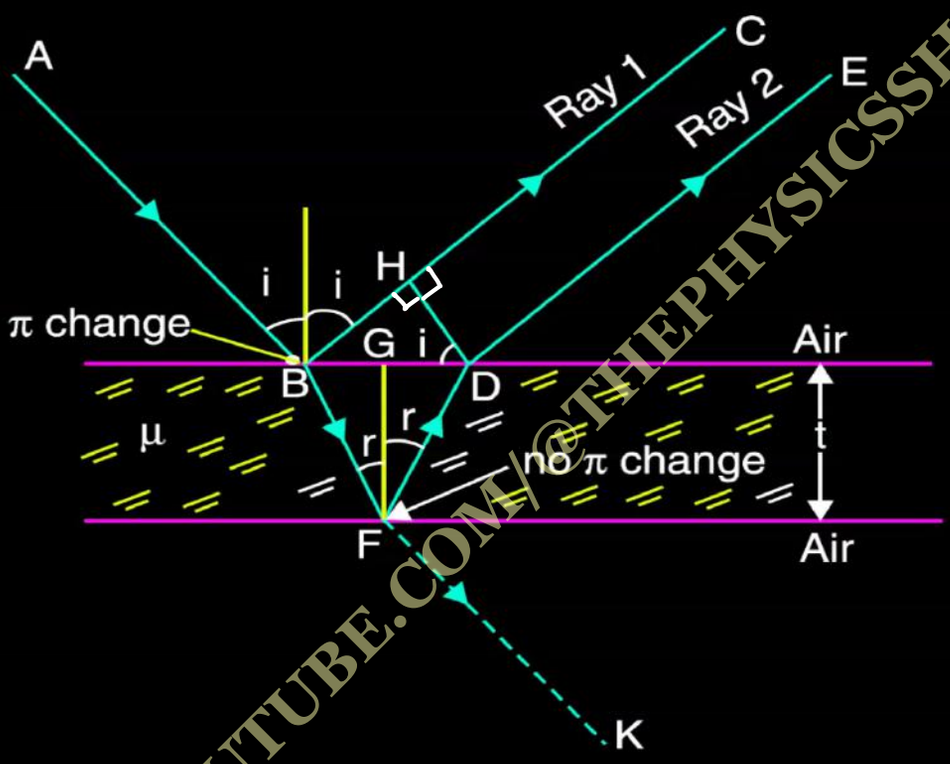
are refractive indices of incidence and refraction media and E_i is the incident electric field Amplitude.

→ Intensity $\propto E^2$ thus for air glass interface, reflected intensity $I_r = \left(\frac{1.52-1}{1.52+1}\right)^2 \times I_i \approx 4\% I_i$.

Thus only about 4% light is reflected.



Optical Path Difference In Thin Film Interference:



BF = t/cos r
BG = t.tan r
BH = 2t.tan r . sin i
$\Delta = 2\mu t \cos r - \lambda/2$

Consider the thin film interference diagram as shown above. Here we need to find optical path difference between paths BFD and BH (after that both rays travel equal paths all the way up to the eye of human observer). Now in the above figure BF = FD = t sec r while BG = GD = t tan r thus BH = BD sin i = 2t tan r sin i. Thus optical path difference (not considering any π shift):

$$\Delta x = 2\mu t \sec r - 2t \tan r \sin i \quad \text{also } \sin i = \mu \sin r$$

thus $\Delta x = 2\mu t(\sec r - \tan r \sin r) = 2\mu t\left[\frac{1}{\cos r} - \frac{\sin^2 r}{\cos r}\right]$

$$\Rightarrow \boxed{\Delta x = 2\mu t \cos r}$$

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→ Now here π shift is happening at one interface and not at the other interface, to account for this either we can subtract $\lambda/2$ from Δx to get Δx_{net} i.e $\Delta x_{net} = 2\mu t\cos r - \lambda/2$ or we can swap the conditions for maxima and minima.

→ $\Delta x = 2\mu t\sqrt{1-\sin^2 r} = 2\mu t\sqrt{1-\left(\frac{\sin i}{\mu}\right)^2} = 2t\sqrt{\mu^2-\sin^2 i}$

thus $\boxed{\Delta x = 2t\sqrt{\mu^2-\sin^2(i)}}$

→ For near normal incidence, $i \approx 0$ thus $\boxed{\Delta x = 2\mu t}$

→ If $t << \lambda$, practically there is no phase difference due to film thickness, all the phase difference is only due to π shift on reflection. Thus if films are very thin, we do not see any VIBGYOR PATTERN because phase difference for all wavelengths is nearly same (either 0 or π) thus the film will appear dark or bright (if π shifts cancel) with uncoloured colour distribution.

→ If $t >> \lambda$, the optical path difference is so large that an integer multiple of almost every colour shade can be fit in that path difference, thus no particular colour is preferred over another and we do not see VIBGYOR pattern from thick films. (for instance if $t = 40,000$ nm, integer multiples of several wavelength ranging between 400 and 700 nm can be fit in the path difference (i.e $2\mu t = 120,000$ nm).

EXAMPLE: White light (spread over 400 nm to 690 nm) is perpendicularly incident on a thin film of water ($\mu = 1.33$) having a thickness $t = 320$ nm with air on both sides. What wavelength(s) will be most strongly reflected by the film. (Incidence is normal)

Solution: For strong reflection, $2\mu t = (2n+1)\frac{\lambda}{2}$ ∵ there is a π shift at first incidence.

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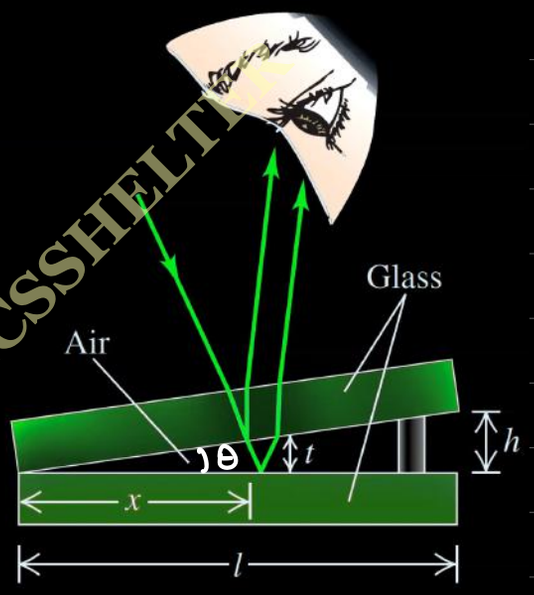
Thus $\lambda = \frac{4\mu t}{2n+1} = \frac{4 \times 1.33 \times 320\,nm}{(2n+1)} = \frac{1702.4}{(2n+1)}$ nm.

The only wavelength that satisfies this in visible range is for $n=1$.
Thus $\lambda = 1702.4/3 = 567.5$ nm. (Greenish Yellow).

→ The light that is most strongly reflected is also most weakly transmitted.

INTERFERENCE FROM THIN WEDGE:

Consider the wedge formed between two microscope slides as shown in adjacent figure. A wedge can be thought of as a thin film of variable thickness. Here also if t is the local thickness we can write path difference as $\Delta x = 2\mu t \cos r$ [provided t is varying very gradually]. Now here there is π shift from lower reflection while no π shift from upper reflection thus conditions for maxima and minima will be swapped. If θ is the wedge angle then we can write $t = x\theta$, also if incidence is nearly normal then $\Delta x = 2\mu t$. Thus for dark fringes:



$$\Delta x = 2\mu t = 2\mu x\theta = n\lambda \Rightarrow \boxed{x = \frac{n\lambda}{2\mu\theta}}$$ ← Dark fringe location.

Now fringe width is the spacing between successive dark bands thus, fringe width here is $\boxed{\beta = \frac{\lambda}{2\mu\theta}}$

→ Note that here $x=0$ is location of dark band because of π shift.

→ In case of air wedge, $\mu = 1$, and $\beta = \lambda/2\theta$.

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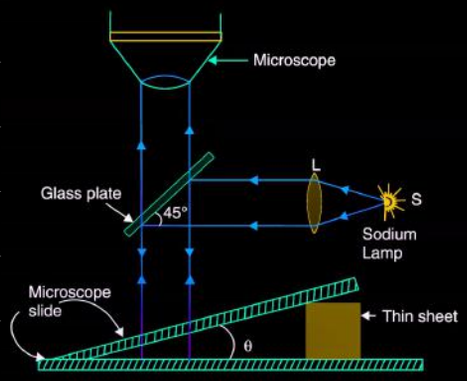
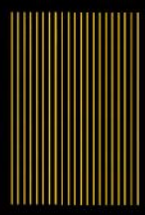
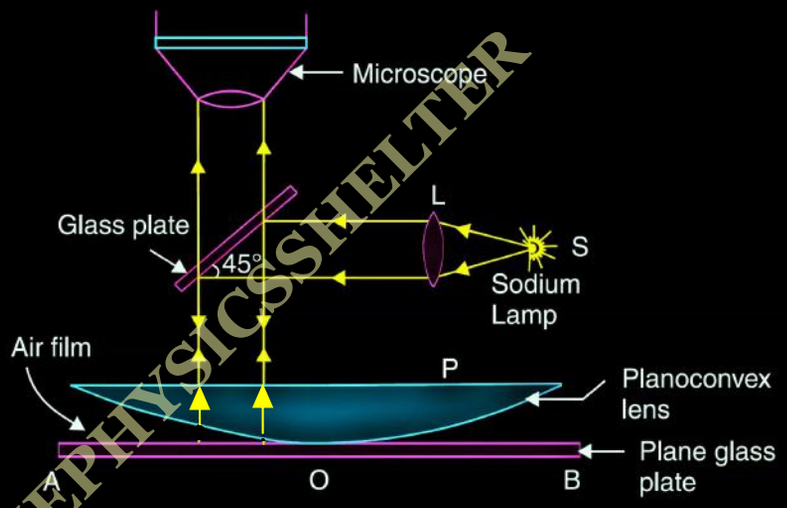
→ The experimental set up for air wedge experiment is shown in adjacent figure. Here the interference fringes are parallel line (equally spaced dark and bright fringes).





NEWTON'S RINGS:

Just like the case of interference with thin wedge, a thin film of a refracting medium can be generated in the gap between a plano convex lens and a glass plate. In principle this set up is similar to the wedge interference set up except that here rate of increase of film thickness increases (θ increases) as one moves radially outwards. Also because of circular symmetry, here we get circular fringes.



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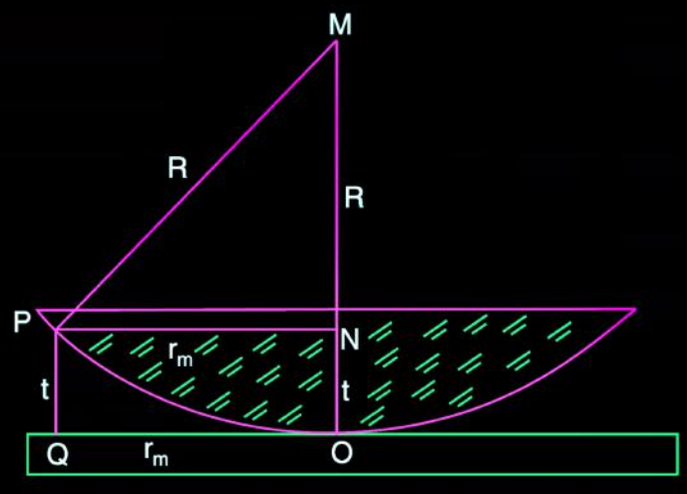
CALCULATING RADII OF DARK FRINGES:

Here also we consider near normal incidence. Thus if local film thickness is t, the effective path difference is $2\mu t$ not accounting for a net π shift. Howeveuer there will be a π shift at one of the reflections thus conditions for maxima and minima will be swapped. Thus for dark fringes,

$2\mu t = n\lambda$ ①



Now considering right ΔMPN,

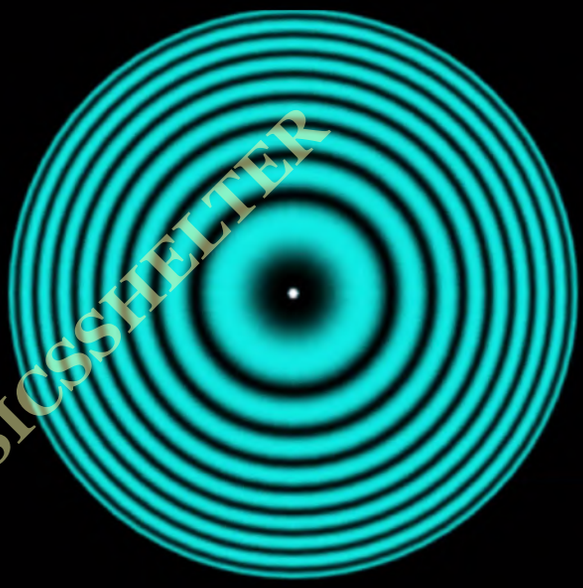
$R^2 - (R-t)^2 = r_m^2$ where r_m is the radius of a dark fringe.

Thus $r_m^2 = 2Rt - t^2$ but $t << R$ typically thus:

$$r_m^2 \approx 2Rt = 2R \times \frac{n\lambda}{2\mu} = \frac{n\lambda R}{\mu} \quad \text{thus} \quad \boxed{r_m = \sqrt{\frac{n\lambda R}{\mu}}}$$

Now radii of different rings can be found by setting $n = 1, 2, 3 \ldots$

→ As one moves out from the centre, the rings get closer and closer and beyond a certain radius, the rings appear indistinguishable because of the restrictions of resolving limit of the human eye.



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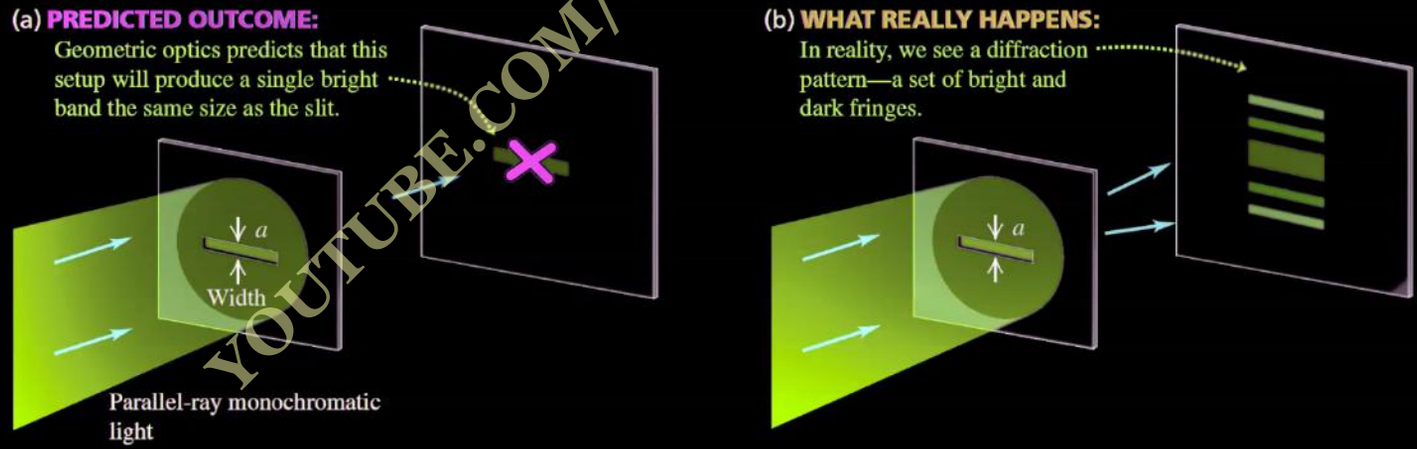
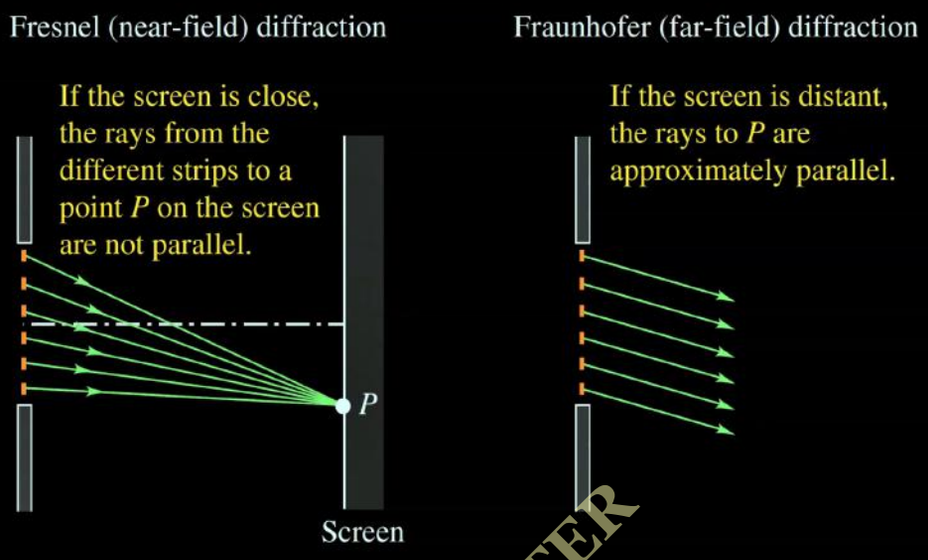
Single Slit Diffraction:



When light is made to pass through a small aperture, by logic of geometrical optics we just expect a single bright strip however actually we see a pattern of bright and dark fringes which can be explained in terms of wave optics.

FRAUNHOFFER Vs FRESNEL DIFFRACTION:



A single slit can be thought of as a collection of infinite number of infinitesimal slits. Depending on the location of the screen, the rays coming from these "slits" may be considered as nearly parallel (if screen is sufficiently far away) or may not be considered as parallel (when screen is not very far). The mathematics of two types of diffraction patterns is different. When the rays can be considered nearly parallel, we call the pattern as Fraunhoffer diffraction and the other case is called Fresnel diffraction. For JEE our discussion will be limited only to Fraunhoffer diffraction patterns.

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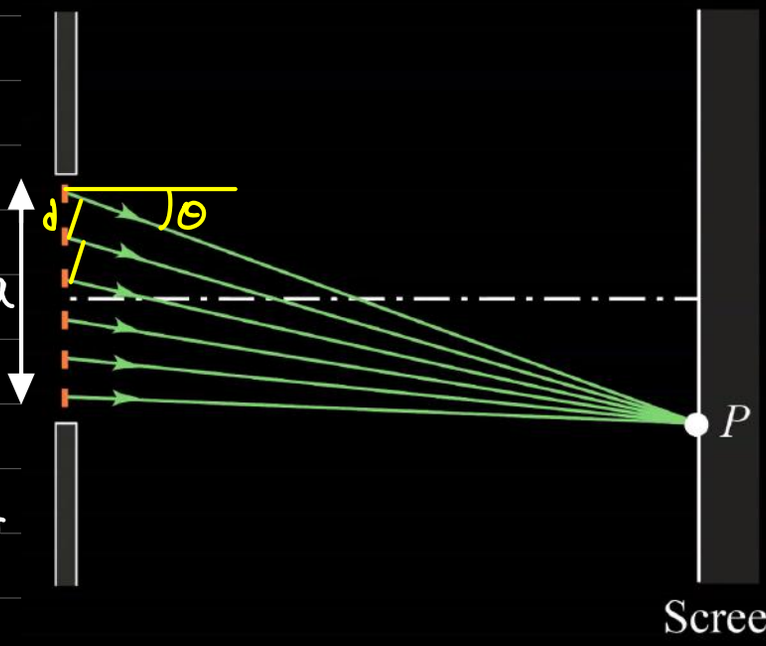
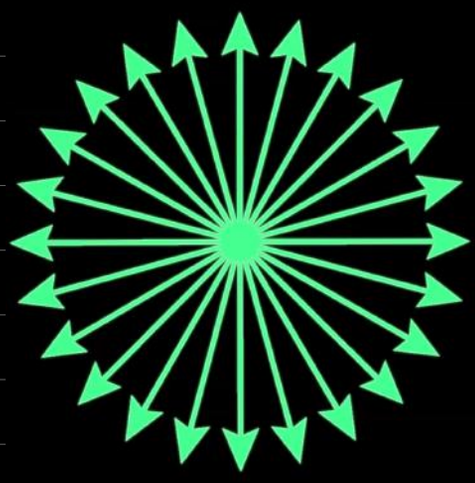
LOCATING MINIMA IN FRAUNHOFFER DIFFRACTION FROM SINGLE SLIT:



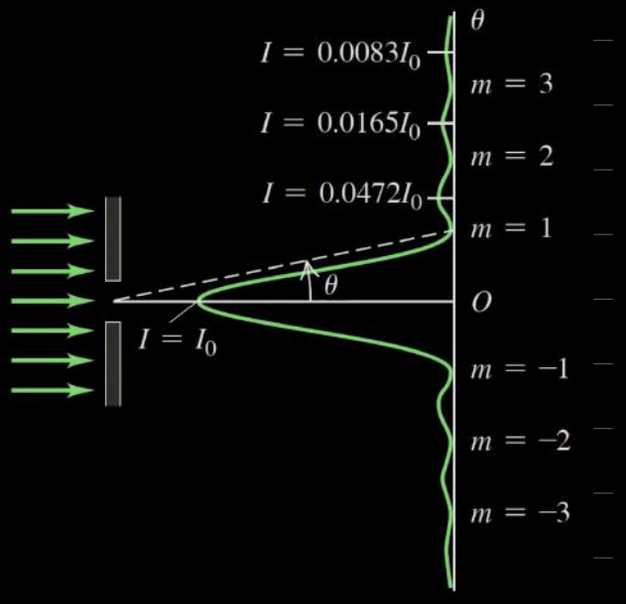
Let us mentally divide the slit in to a very large number of equal parts (say l mol parts) mentally. Further let separation between successive slits be d and rays are incident at an angle θ with normal to the screen. The path difference between rays from successive slits will be $d\sin\theta$ each. Thus now we have l mol equally spaced phasors (each leading the previous slit by an amount $\frac{2\pi}{\lambda}.d\sin\theta$).



Now imagine I_{mol} phasors arranged coinitially. If the phase difference between the first and the last phasor is 2π, then we have infinite no of infinitesimal phasors spread uniformly over a circle, thus giving a net zero resultant. Similarly the resultant would be zero if the phasors were uniformly spread over an angle 4π, 6π or in general $2m\pi$. Thus for a minimum, the phase difference between topmost phasor and bottom-most phasor must be $2m\pi$. Now if a is the width of the slit then path difference between topmost phasor and bottommost phasor is $a\sin\theta$. Thus for minima, $\frac{2\pi}{\lambda}\cdot a\sin\theta = 2m\pi$ or

$$\boxed{\sin\theta = \frac{m\lambda}{a}} \quad m = \pm1, \pm2, \pm3\ldots \; m \neq 0$$ ← Angular position of minima.

→ For $m=0$, All phasor are in same phase (distributed in a 0° angle) thus we have central maximum for $\theta = 0$.



Angular Width Of Central Maximum:

The extent of central maximum is considered as stretching between first order minima on either sides of central maximum. If θ is small then $\sin\theta = \theta$ thus angular position of minima is $\theta = m\lambda/a$ thus angle between first order minima on either side : $2\theta = 2\lambda/a$ thus angular width of central maximum, $\boxed{\omega = 2\lambda/a}$

Location Of Maxima: Maxima can be located using the calculus approach, however it is fairly complicated. For simplicity we may assume maxima to be located at mid points of successive minima. Thus for maxima, $a\sin\theta = \pm(2n+1)\lambda/2$ or $\boxed{\sin\theta = \pm(2n+1)\frac{\lambda}{2a}}$, $n = 1, 2, 3\ldots$

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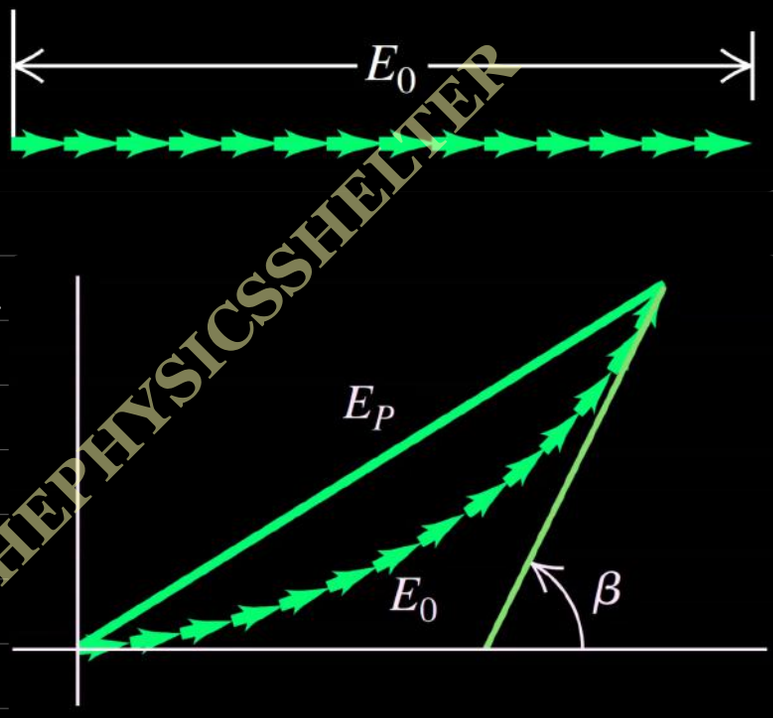
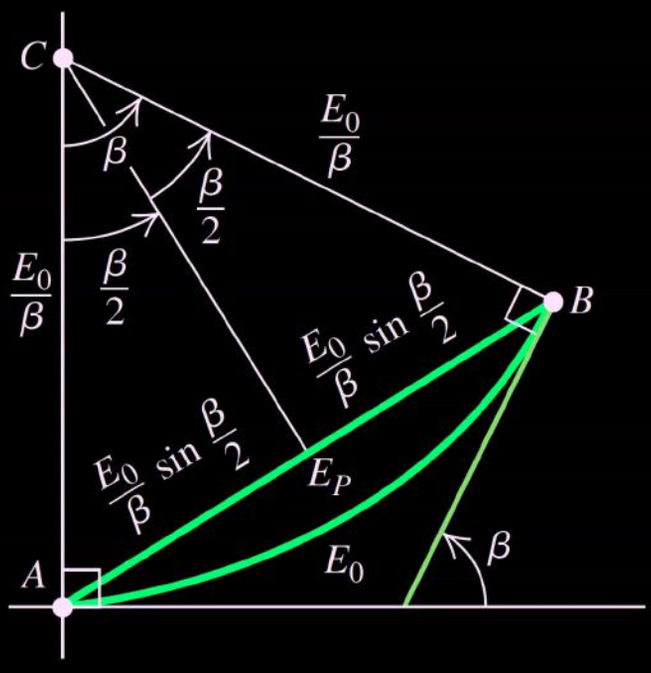
→ Note that minima occurs for a phase difference of 2π and the extent of (central) maximum is taken up to first order minima, ∴ no maximum is counted for phase difference of π between top and bottom phasors. The phase differences considered for maxima are 3π, 5π, 7π etc. ∴ $n \neq 0$ when we write for maxima $\sin\theta = \pm (2n+1)\frac{\lambda}{2a}$.

Relative Intensity In Single Slit Diffraction:

Let E_0 be the resultant Amplitude of all the phasors at the central maximum (Here algebraic sum is same as vector sum because all phasors are aligned together). Now let E_P be the phasor sum of all the phasors when phase difference between top phasor and bottom phasor is β. Then the situation is shown in the second figure. Here arc length is still E_0 (algebraic sum) while chord length represents vector sum E_P. Now for a circular arc, $l = R\theta$, Thus, as seen from the third figure, the radius of the arc must be E_0/β while the chord length is



$$E_P = \frac{2E_0}{\beta}\sin(\beta/2) = \frac{E_0 \sin(\beta/2)}{(\beta/2)}$$



Now intensity is proportional to the square of amplitude thus if I_0 is intensity of central maximum, the intensity at a general point is:

$$I_P = I_0 \left(\frac{\sin(\beta/2)}{\beta/2}\right)^2 \quad \text{where} \quad \beta = \frac{2\pi}{\lambda}\cdot a\sin\theta$$

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→ If we wish to locate exact angular position of maximum, we can now put $dI_P/d\theta = 0$, However it leads to a transcendental equation not solvable by hand (can be solved numerically with help of a computer or calculator).

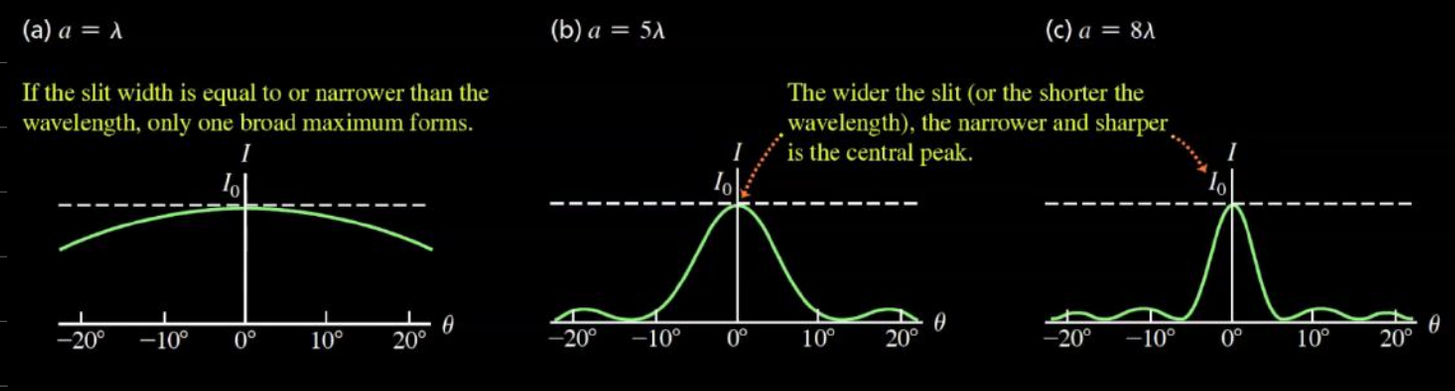
→ Relative intensities of successive maxima excluding the central one will be $\frac{1}{9} : \frac{1}{25} : \frac{1}{49}$ etc this is because for first maxima the phasors are distributed in 3 semicircles, for second 5 semicircles and so on. Thus after a few bright bands the pattern practically vanishes.

→ Relative intensity of first maxima (relative to central) can be found by putting $\beta = 3\pi$. Thus $I_1 = I_0\left[\frac{\sin(3\pi/2)}{(3\pi/2)}\right]^2 = \frac{4I_0}{9\pi^2} \approx 0.045\, I_0$

(Actually it is $0.047\, I_0$, however we approximated maximum at bisector of two minima). Using the bisector approach, the intensity of side maxima come out to be $0.045\, I_0$, $0.0162\, I_0$ and $0.0083\, I_0$ while actually they are $0.0472\, I_0$, $0.0165\, I_0$ and $0.0083\, I_0$, so our approximation is fairly good.

→ Approximate intensity of m^{th} side maximum using our formula comes out to be, $I_m = \frac{I_0}{[(m+\frac{1}{2})\pi]^2}$

→ Recall that for minima, $a\sin\theta = n\lambda$ or $\sin\theta = n\lambda/a < 1$ thus $n < a/\lambda$. Thus as a increases, the number of minima increases, and when $a < \lambda$, there is only one broad maximum formed. (See the figure below):



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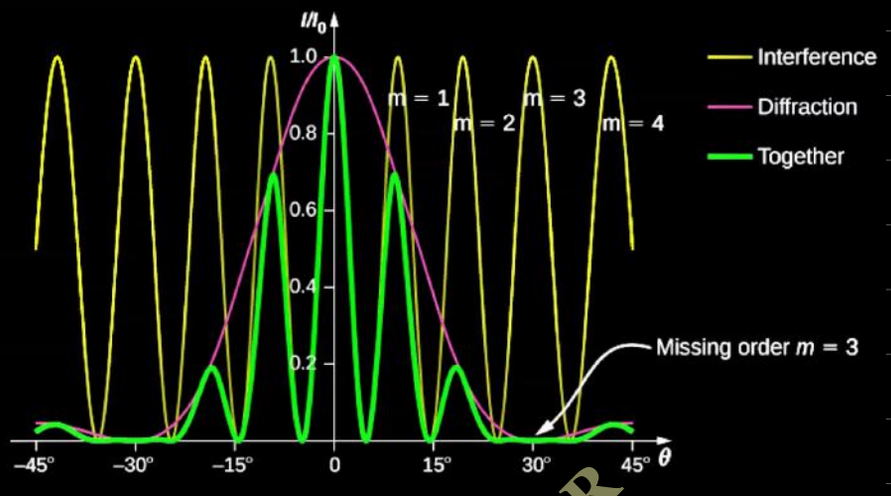
Intensity In A Double Slit Diffraction Pattern:

When we studied Young's double slit experiment, we ignored the diffraction effects. Which is a reasonable assumption if the slits are very very small. (Recall from previous section that when slits are very small, no minima are formed (and) i.e $a << \lambda$). However if slits are wide enough, diffraction effects from slits cannot be ignored.



→ The diffraction pattern of two slits of width a each that are separated by a distance d is the interference pattern of two point sources multiplied by the diffraction pattern of slit of width a.

→ Due to diffraction effects, certain fringes in YDSE may go missing because even though according to YDSE there should be a maximum there, however due to diffraction, there is a minimum there.

→ For double slit diffraction, $I = I_0 \cos^2\left(\frac{\Delta\phi}{2}\right) \cdot \left[\frac{\sin(\beta/2)}{\beta/2}\right]^2$

where $\beta = \frac{2\pi}{\lambda} a \sin\theta$ and $\Delta\phi = \frac{2\pi}{\lambda} d \sin\theta$. Here I_0 is maximum intensity (centre of the screen).

Example: Suppose in YDSE, $a = 2\lambda$ and $d = 6\lambda$, calculate the intensity of the first order maximum relative to the central maximum.

Solution: $I = I_0 \cos^2(\Delta\phi/2)\left[\frac{\sin(\beta/2)}{(\beta/2)}\right]^2$ Now at first order maximum, $\Delta\phi = 2\pi$.

$d \sin\theta = \lambda \Rightarrow \sin\theta = \lambda/d = 1/6$, now $\beta = \frac{2\pi}{\lambda} \cdot a \sin\theta = \frac{2\pi}{\lambda} \times 2\lambda \times \frac{1}{6}$

$\Rightarrow \beta = 2\pi/3$. Thus $I = I_0 \times 1 \left(\frac{\sin(\pi/3)}{\pi/3}\right)^2 = 0.684\, I_0$ Ans.

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EXAMPLE: Suppose in YDSE slits of width 0.02 mm are separated by 0.2 mm. If the slits are illuminated by monochromatic light of wavelength 500 nm, how many bright fringes are observed in central peak of diffraction pattern?

Solution: Angular position of first diffraction minimum is $\sin\theta = \lambda/a$ i.e $500\times10^{-9}/0.02\times10^{-3} = 2.5\times10^{-2}$, Now for n^{th} order maximum, $d\sin\theta = n\lambda \Rightarrow n = \frac{d\sin\theta}{\lambda} = \frac{0.2\times10^{-3}}{500\times10^{-9}}\times 2.5\times10^{-2} = 10$

Thus 10^{th} order maximum overlaps with diffraction minimum, so this fringe will go invisible. Thus number of fringes within diffraction maximum: $N = 9\times 2+1$ i.e 19 fringes Ans.

FRESNEL DISTANCE: The screen distance for which half width of central maximum becomes equal to the size of the slit is called Fresnel distance, mathematically,

$$\boxed{z_F = a^2/\lambda} \quad \left(\because \frac{\lambda}{a} z_F = a \Rightarrow z_F = a^2/\lambda\right).$$

At screen distances much smaller than z_F, spreading due to wave optics is smaller compared to the size of the beam, for distances much greater than z_F, spreading due to diffraction dominates over the geometric optics effects.

→ Alternatively we can define fresnel distance as the distance at which increase in spreading of light is equal to the aperture. (i.e width of central maximum is 2a (i.e a+a))

YDSE WITH SLITS OF UNEQUAL WIDTH: With slits of unequal width, for an accurate analysis diffraction effects should be considered however in popular literature, individual intensities due to each slit are taken as simply directly proportional to slit width. (Although it is wrong, but that's what most question setters expect you to do).

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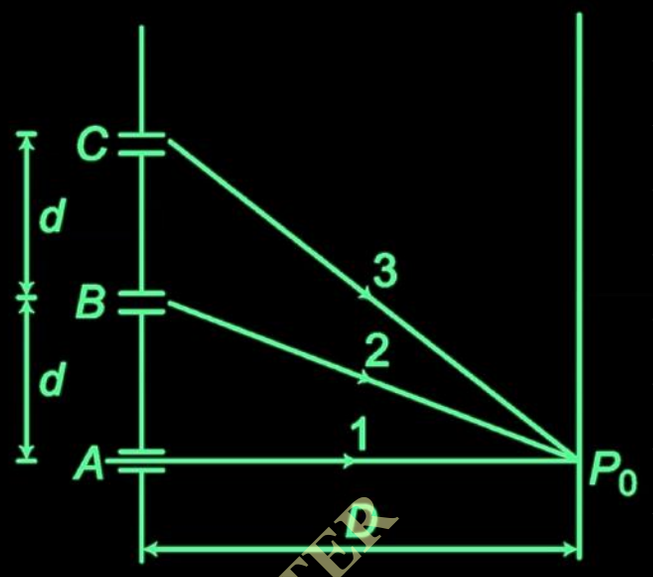
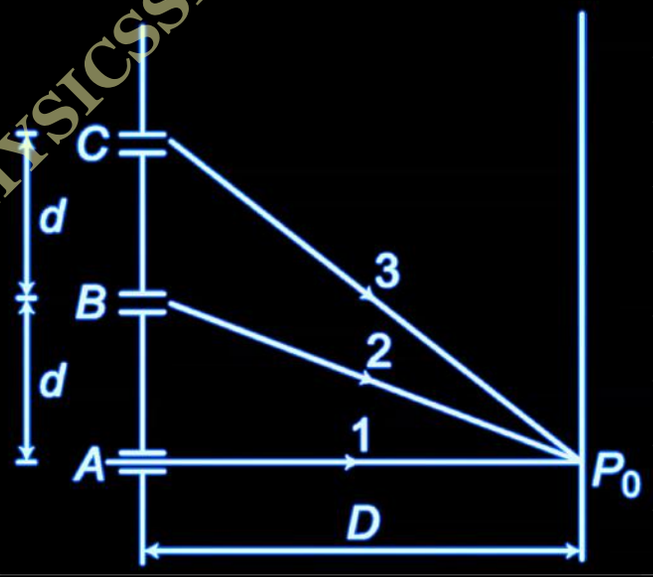
Multiple Slit Interference:

To find intensity due to multiple slits, we must properly figure out the phase of each wave relative to the longest optical path (most lagging) and then add the amplitudes vectorially considering the proper phase of each wave. We can use this technique to find out intensity due to any number of waves.

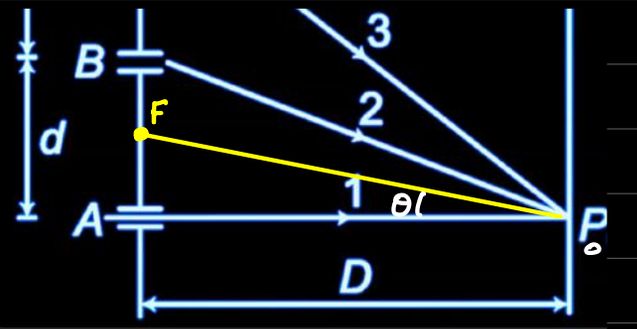


Example: Figure shows three equidistant slits illuminated by a monochromatic parallel beam of light. Let $BP_0 - AP_0 = \lambda/3$ and $D >> \lambda$ (a) Show that $d = \sqrt{2\lambda D/3}$ (b) Show that intensity at P_0 is three times the intensity due to any of the three slits individually. (Also $D >> d$)



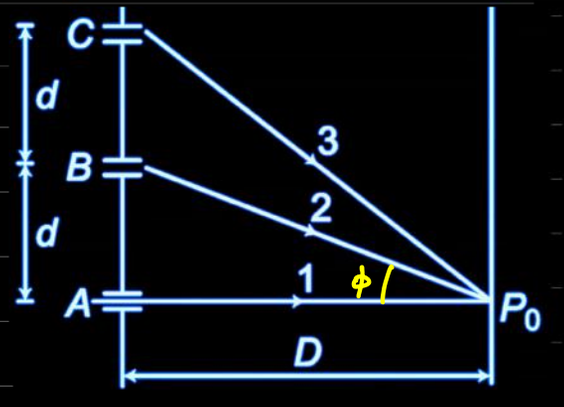
Solution: (a) Let F be the mid point of AB. Then $BP - AP = d \sin\theta \approx d \tan\theta$ ($\because \theta$ is small) $= d \times \frac{d}{2D} = \lambda/3$

$$\Rightarrow \quad d = \sqrt{\frac{2D\lambda}{3}} \quad \text{QED.}$$



Similarly $CP_0 - AP_0 = 2d \sin\phi \times 2d \tan\phi$

$$= 2d \times \frac{d}{D} = \frac{2d^2}{D} = \frac{4\lambda}{3}$$



Now λ corresponds to a phase lag of 360° thus $\frac{\lambda}{3}$ corresponds to a lag of 120° and $4\lambda/3$ corresponds to a lag of 480° which

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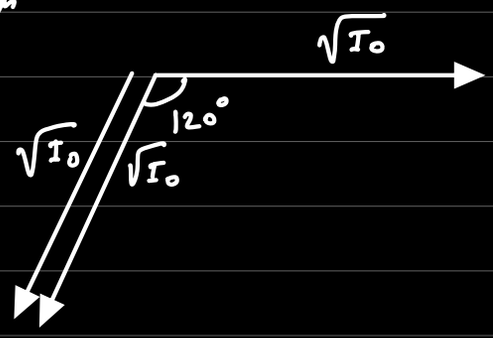
is equivalent to a lag of 120°. Thus the relevant phasor diagram for intensity calculation is shown in the adjacent figure. (∵ Intensity $\propto A^2 \Rightarrow A \propto \sqrt{I}$). Thus required resultant intensity:



$$\left(2\sqrt{I_0}\right)^2 + \left(\sqrt{I_0}\right)^2 + 2\sqrt{2I_0}\sqrt{I_0}\cos 120°$$

$$= 5I_0 - 2I_0 = 3I_0$$ (Here I_0 is individual intensity due to each slit).

Thus $I = 3I_0$ QED.

EXAMPLE: IRODOV 5.66:

A certain oscillation results from the addition of coherent oscillations of the same direction $\xi_k = a\cos[\omega t + (k-1)\phi]$, where k is the number of the oscillation ($k = 1, 2, \ldots, N$), ϕ is the phase difference between the kth and $(k-1)$th oscillations. Find the amplitude of the resultant oscillation.

Solution: Here we wish to get the phasor sum of given N phasors, we can do it either geometrically or expressing the phasors as complex numbers and taking the real part. We will follow the complex numbers approach. Now in phasor domain, $a\cos(\omega t + \phi)$ is written as $ae^{j(\omega t + \phi)} = ae^{j\omega t} \cdot e^{j\phi}$. Thus our required phasor sum becomes:

$$ae^{j\omega t}\left[1 + e^{j\phi} + e^{2j\phi} + \ldots e^{(N-1)j\phi}\right]$$

This sum of a GP. Thus:

$$\tilde{E} = ae^{j\omega t}\left[\frac{1 - e^{Nj\phi}}{1 - e^{j\phi}}\right] \Rightarrow E_{max} = a\left[\frac{(1 - e^{Nj\phi})(1 - e^{-Nj\phi})}{(1 - e^{j\phi})(1 - e^{-j\phi})}\right]^{1/2}$$

$$\Rightarrow E_{max} = a\left[\frac{1 + 1 - e^{-Nj\phi} - e^{Nj\phi}}{1 + 1 - e^{j\phi} - e^{-j\phi}}\right]^{1/2} = a\left[\frac{2 - 2\cos N\phi}{2 - 2\cos\phi}\right]^{1/2}$$ (∵ $|z| = (z\bar{z})^{1/2}$)

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$\Rightarrow E_{max} = a\left[\sin(N\phi/2) / \sin(\phi/2)\right]$ Ans.

$\longrightarrow$ This can be used to find intensity due to N equidistant slits, using $I \propto E_{max}^2$

EXAMPLE: IRODOV: 5.67:

A system illustrated in Fig. 5.12 consists of two coherent point sources 1 and 2 located in a certain plane so that their dipole moments are oriented at right angles to that plane. The sources are separated by a distance d, the radiation wavelength is equal to λ. Taking into account that the oscillations of source 2 lag in phase behind the oscillations of source 1 by ϕ ($\phi < \pi$), find:

(a) the angles θ at which the radiation intensity is maximum;

(b) the conditions under which the radiation intensity in the direction $\theta = \pi$ is maximum and in the opposite direction, minimum.

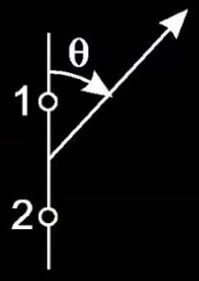

Fig. 5.12

Solution: Consider a general angle θ as shown in the problem figure. If there we no initial phase lag, due to path difference alone, wave from 2 lag behind 1 by a phase of $\frac{2\pi}{\lambda} \cdot d\cos\theta$ Thus, net phase lag becomes $\Delta\phi = \frac{2\pi}{\lambda} d\cos\theta + \phi$.

Now for maximum intensity, $\Delta\phi = 2n\pi$, thus, $\frac{2\pi}{\lambda} d\cos\theta + \phi = 2n\pi$

$$\Rightarrow \cos\theta = \left[\frac{\lambda}{d}\left[\frac{2n\pi - \phi}{2\pi}\right]\right] = \left[\frac{\lambda}{d}\left(n - \frac{\phi}{2\pi}\right)\right] \text{ Ans.} \qquad [n \in I]$$

Now a phase lag of ϕ is equivalent to an additional path length of $\frac{\phi}{2\pi}\lambda$.

Effective path difference for $\theta = 0°$: $\Delta x_1 = d + \frac{\phi}{2\pi}\lambda = (2n+1)\frac{\lambda}{2}$ (1) (∵ destructive).

Effective path difference for $\theta = \pi$, $\Delta x_2 = d - \frac{\phi}{2\pi}\lambda = n\lambda$ (2)

(1) − (2) $\Rightarrow \frac{\phi\lambda}{\pi} = \left(n - m + \frac{1}{2}\right)\lambda$ or $n - m + \frac{1}{2} - \frac{\phi}{\pi} = 0$

Now ∵ $0 < \phi/\pi < 1 \Rightarrow -\frac{1}{2} < (n-m) < \frac{1}{2} \Rightarrow n - m = 0 \Rightarrow \phi = \pi/2$ Ans.

Now putting $\phi = \pi/2$ in (2), $d = \left(m + \frac{1}{4}\right)\lambda$ Ans.

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Example: Irodov 5.68:

A stationary radiating system consists of a linear chain of parallel oscillators separated by a distance d, with the oscillation phase varying linearly along the chain. Find the time dependence of the phase difference $\Delta\phi$ between the neighbouring oscillators at which the principal radiation maximum of the system will be "scanning" the surroundings with the constant angular velocity ω.

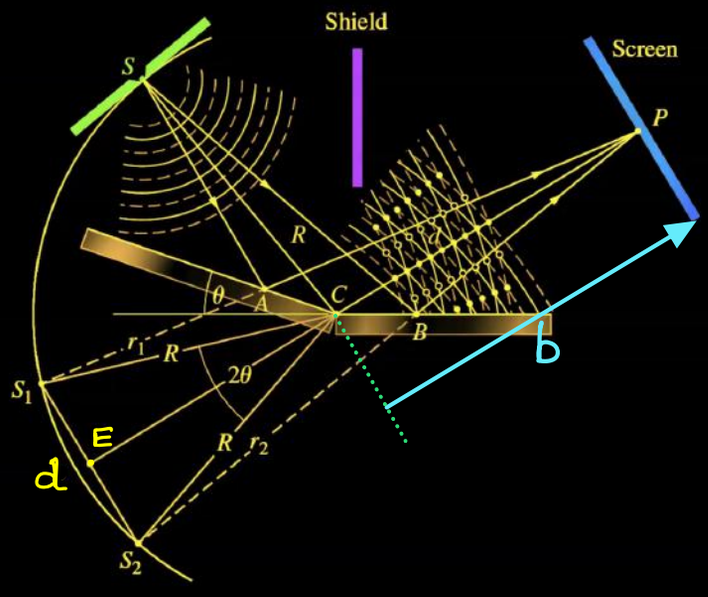
Solution: Let phase difference between neighboring oscillators be $\Delta\phi$. Then for constructive interference, $\frac{2\pi}{\lambda} d\cos\theta + \Delta\phi = 2n\pi$ ①

Now for scanning, θ changes with time as $\theta = \omega t + \delta$, thus:

$$\frac{2\pi}{\lambda} d\cos(\omega t + \delta) + \Delta\phi = 2n\pi \quad \text{or} \quad \Delta\phi = 2\pi\left[n - \frac{d}{\lambda}\cos(\omega t + \delta)\right] \text{ Ans.}$$

Fresnel Double Mirror (Irodov 5.71)

The figure shows interference experiment with Fresnel mirrors. The angle between mirrors, $\theta = 12'$. The distances from mirrors intersection line to the narrow slit S and the screen are equal to $R = 10.0$ cm and $b = 130$ cm respectively. The wavelength of light $\lambda = 0.55\ \mu$m.



Find: (a) Fringe width and number of possible maxima

(b) Shift of interference pattern when slit is displaced by $S = 1$ mm along arc of radius R with centre at point C

(c) At what maximum width w of the slit the interference fringes

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on the screen are still observed sufficiently sharp.

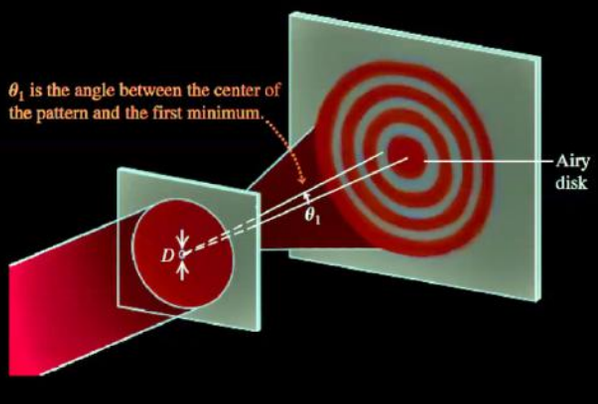
Solution: From the figure, if angle between mirrors is θ, angle between the two images is 2θ. Thus, $d = 2R\theta$, Now θ is very small thus $CE \approx R$ and $D = CE + CP = R + b$. Thus, fring width, $\beta = \lambda D/d = \lambda(R+b)/2R\theta = 1.1\text{mm}$ Ans. Now interference is obtained between lowest ray from S_1 and highest ray from S_2 i.e in an angular width 2θ or linear width $2b\theta$, i.e $b\theta$ on either side of central maximum. Thus no of maxima: $n = \text{GIF}(b\theta/\beta) \times 2 + 1 = 8 + 1 = 9$ Ans (a)

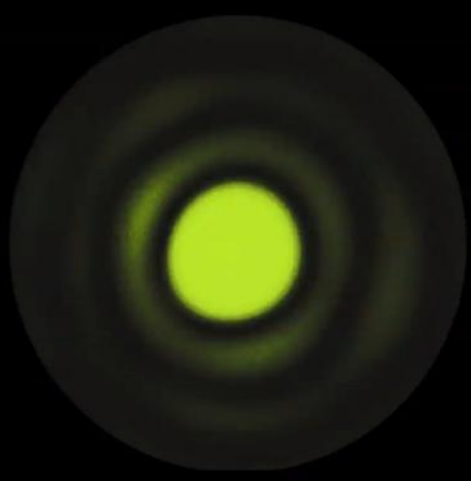
(b) Now when slit is shifted by δ angular shift of slit, $\phi = \delta/R$
Now S_1 and S_2 must rotate by an additional ϕ and so also their bisector. Thus pattern shifts by $\Delta y = b\phi = b\delta/R = 13\text{mm}$ Ans (b)

(c) For finding maximum allowable width of slit, we can imagine the slit to be two point sources at its ends separated from each other by some distance δ. If δ is such that the fringe pattern from one end is shifted by $\beta/2$ from the pattern from the other end, then the maxima from one end will overlap with minimum from the other end, and the pattern will become indistinguishable. Thus, $\Delta y = \beta/2$ or $\frac{b\delta}{R} = \left(\frac{\lambda(R+b)}{2R\theta}\right)\cdot\frac{1}{2}$

$$\Rightarrow w = \delta = \frac{\lambda(R+b)}{4b\theta} = 42.4\mu\text{m}$$ Ans.

FRAUNHOFFER DIFFRACTION FROM CIRCULAR APERTURE :





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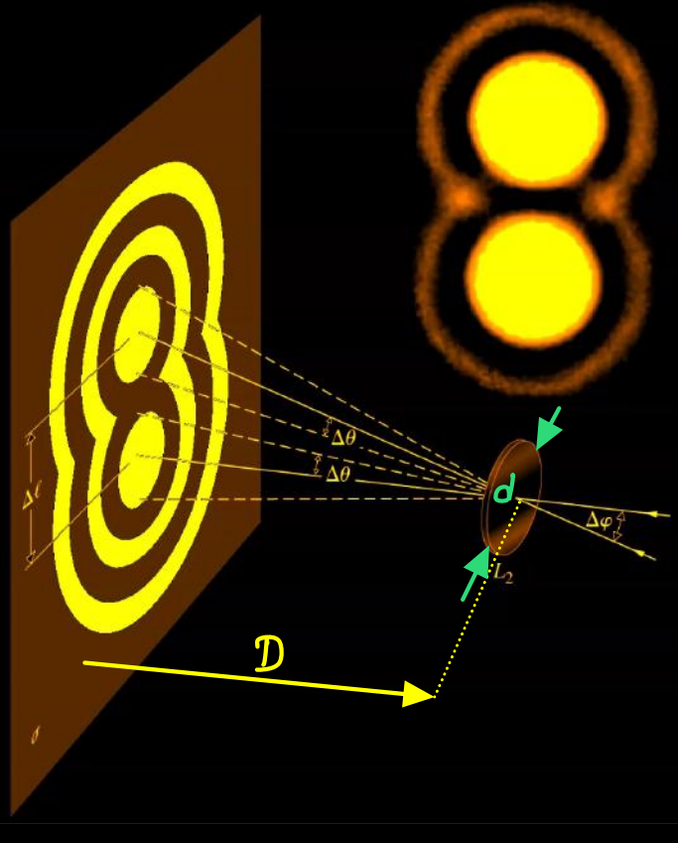
In this section we consider the diffraction by circular apertures, the analysis of such apertures is mathematically complicated though from point of view of physics, it is similar to that of slit diffraction. As clear from the images above it can be seen that major portion of light is concentrated in a circular region and then we have very faint concentric dark and light bands (barely visible). The central bright circle is called Airy disc and it subtends an angular radius θ at the centre of the circular aperture such that, $\boxed{\sin\theta = \frac{1.22\lambda}{d}}$ where d is the diameter of the aperture. The result for angular radius is also applicable to lenses of circular aperture. Thus if rays are incident from infinity on a lens it forms a bright spot of radius $R = f\theta = \frac{1.22\lambda f}{d}$ in the focal plane.

The parallel rays coming from different directions will produce different circular spots in the focal plane whose centres will lie along the ray passing through centre of the aperture. Depending upon the angular separation between distant objects, their airy discs may either significantly overlap or not overlap. If there is too much overlap of the airy discs, it becomes hard to distinguish the images of the distant sources. If D is the screen distance and d the size of the aperture and $\Delta\phi$ the angle between distant objects as seen from the aperture, then the distance between centres of two airy discs, $\Delta l = D\Delta\phi$, while radii of



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of each airy disc is $R = \frac{1.22\lambda D}{d}$. Based on the relationship between radii of airy discs and centre to centre separation between two airy discs, Lord Rayleigh stated a criterion for considering the two images to be resolved or not resolved. This will be discussed in the next section.

RAYLEIGH'S CRITERION:

According to Rayleigh's criterion, two images in fraunhofter diffraction from circular apertures are said to be resolved if centre to centre distances of Airy discs is more than the radii of the discs. Thus using formulae from previous section, for airy discs to be resolved, $\Delta l > R$ or $D\Delta\phi > \frac{1.22\lambda D}{d}$

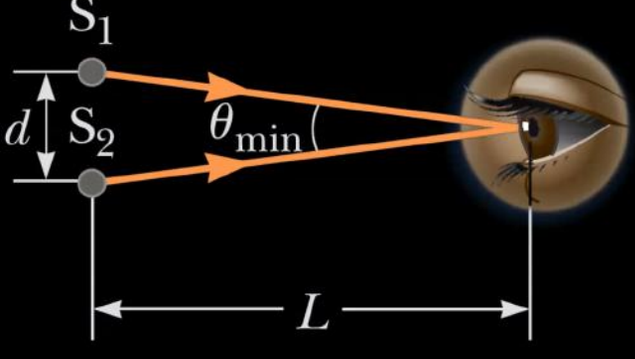
or $\boxed{\Delta\phi > 1.22\lambda/d}$ (1) ← Rayleigh's criterion.

Thus for two images to be resolved in fraunhofter diffraction, their angular separation should be atleast:

$$\boxed{\Delta\phi_{min} = \frac{1.22\lambda}{d}}$$

If in equation (1) if there is equality, the objects are said to be barely resolved and if $\Delta\phi$ is even less than that, the images are said to be unresolved.

EXAMPLE: Light of wavelength 500 nm near the centre of visible spectrum enters a human eye. Although pupil diameter varies from person to person, we may assume a daytime diameter of 2 mm (a) Estimate the limiting angle of resolution



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for this eye, assuming it is limited only by diffraction.

(b) Determine the minimum separation distance d between two point sources that the eye can distinguish if the point sources are at a distance $L = 25$ cm from the observer.

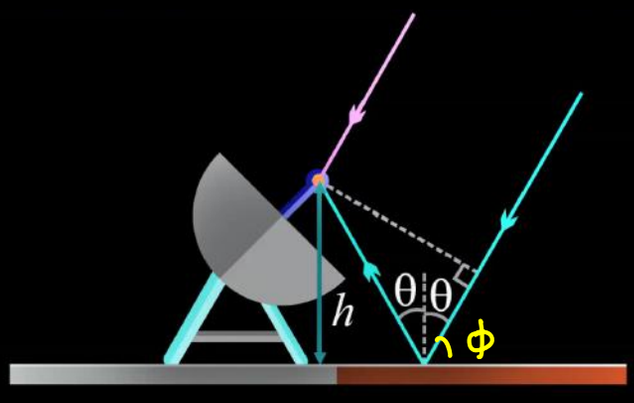
Solution: $\theta_{min} = 1.22\lambda/d = 1.22 \times 500\,nm / 2mm = 305 \times 10^{-6}$ rad $\approx 3 \times 10^{-4} \times \frac{180}{\pi} \times 60$ minutes ≈ 1 minute of arc Ans (a)

(b) Now min required linear distance, $\Delta l_{min} = L\,\theta_{min} = 25 \times 305 \times 10^{-6}$ cm ≈ 0.008 cm $= 8 \times 10^{-3}$ cm Ans (b).

EXAMPLE : PATHFINDER MCQ 10:

In the middle of a calm lake, a radio receiver is installed on a pole at a height 4.0 m above water level. The receiver is used to track radio signals from a satellite orbiting the earth. As the satellite rises above the horizon, intensity of the signals at the receiver varies periodically. It is maximum, when the satellite is 3° above the horizon and then again when the satellite is 6° above the horizon. Wavelength of the satellite signals is closest to

(a) 24 cm (b) 36 cm
(c) 42 cm (d) 48 cm



Solution: Consider the adjacent figure. The path difference is $h\sec\theta + h\sec\theta\cos 2\theta$

i.e $\Delta x = h\sec\theta\,[1 + \cos 2\theta]$

$\Rightarrow \Delta x = 2h\sec\theta\cos^2\theta = 2h\cos\theta$ also reflected wave undergoes a π shift. In terms of ϕ, $\Delta x = 2h\sin\phi$, Since the condition changes from a maximum to next maximum as ϕ changes from $\phi_1 = 3°$ to $\phi_2 = 6°$, $\therefore$ path difference must have changed by λ in this time. Also since ϕ is very small, we may take $\sin\phi \approx \phi$. Thus: $\Delta x = 2h(\sin\phi_2 - \sin\phi_1)$

$2 \times 4\left[\frac{6\pi}{180} - \frac{3\pi}{180}\right] = \lambda \Rightarrow \lambda = 41.9$ cm ≈ 42 cm Ans (c)

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Example: Pathfinder: Build 34:

A liquid of refractive index μ is poured on a large horizontal clear glass plate to make a thin layer of it. When a broad light beam is projected on the liquid layer at an angle θ with the vertical, an interference pattern is obtained. Due to evaporation, thickness of the liquid layer decreases making changes in the interference pattern. If in a time interval Δt, intensity of reflected light at a point decreases from a maxima to a minima and then increases to a maxima, find suitable expression for the time rate at which thickness of the liquid layer decreases.

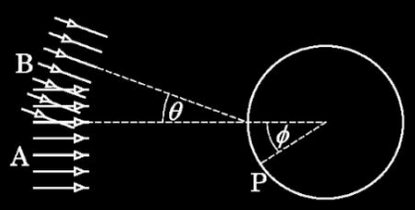
Solution: Path difference $\Delta x = 2\mu t \cos r = 2\mu t \sqrt{1-(\sin i/\mu)^2}$

$\Rightarrow \Delta x = 2t\sqrt{\mu^2 - \sin^2\theta}$, Now let rate of decrease of thickness be r then in time Δt, Δx must change by λ since the condition changes from a maximum to next maximum Thus:

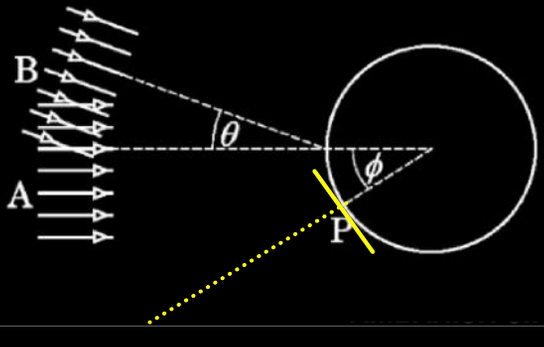
$$\lambda = 2r\Delta t\sqrt{\mu^2-\sin^2\theta} \quad \Rightarrow r = \frac{\lambda}{2\Delta t\sqrt{\mu^2-\sin^2\theta}} \text{ Ans.}$$

Example: Pathfinder: Check 15:

Two broad monochromatic beams A and B of plane coherent waves of the same intensity and wavelength λ propagating at angle θ from each other, simultaneously illuminate a cylindrical screen. Directions of propagation of both the beams are perpendicular to the axis of the screen. Consider a point P on the screen at angular position ϕ from the direction of propagation of the beam A as shown in the figure. Find distance between adjacent interference fringes on the screen near the point P. Assume that the distance between adjacent fringes is much less than the radius of the cylinder.



Solution: A small local patch near point P may be considered nearly flat, then the problem reduces to interference between two parallel beams. The two beams lie on same side of the normal at angles of incidence ϕ and $(\phi+\theta)$. Thus, fringe width is found using the standard result, $\beta = \frac{\lambda}{\sin(\theta+\phi)-\sin\phi}$



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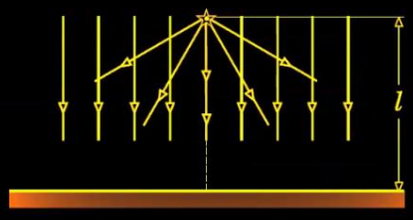
$$\Rightarrow \beta = \frac{\lambda}{2\sin(\theta/2)\cos(\phi + \theta/2)} \quad \text{Ans.}$$

EXAMPLE: PATHFINDER: CHECK 16:

Spherical waves emanated from an isotropic point source of light located at a distance l from a plane screen and a broad parallel beam of light emanated from an extended source simultaneously illuminate the screen. The parallel beam falls on the screen at normal incidence. Both the sources are coherent, emit monochromatic light of wavelength λ and the point source is in phase with a wavefront of parallel beam at its location.



(a) What kind of interference pattern is obtained on the screen?
(b) Find expression for the spacing between the n^{th} and the $(n-1)^{th}$ bright fringe.

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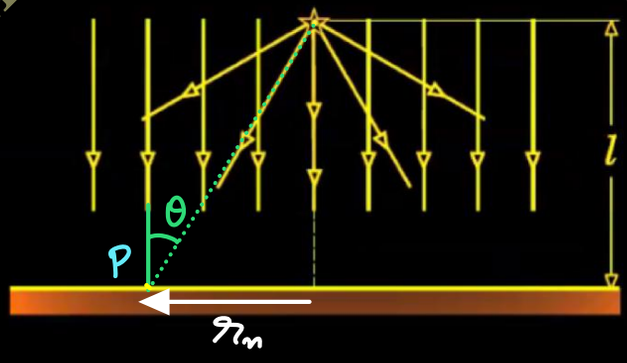
Solution: Due to circular symmetry about axis of the setup, we expect a circular fringe pattern with a bright spot at the centre (since path difference is zero there).

Now consider a point P. The path difference here is $\Delta x = \sqrt{l^2 + r^2} - l$

For n^{th} bright fringe this must be $n\lambda$.

Thus:

$$\sqrt{l^2 + r_n^2} - l = n\lambda \quad (1)$$



also

$$\sqrt{l^2 + r_{n-1}^2} - l = (n-1)\lambda \quad (2)$$

r_n: radius of n^{th} bright ring.

Solving (1) & (2),

$$\Delta x = r_n - r_{n-1} = \sqrt{n\lambda(2l + n\lambda)} - \sqrt{(n-1)\lambda[2l + (n-1)\lambda]} \quad \text{Ans.}$$

If $r_n << l$ then we may use approximation to get the answers in the pathfinder. In that case using binomial approximation,

(1) gives: $l\left[1 + \frac{1}{2} r_n^2/l^2\right] - l = n\lambda$

$\Rightarrow r_n \simeq \sqrt{2n\lambda l}$ similarly $r_{n-1} \simeq \sqrt{2(n-1)\lambda l}$ thus:

$\Delta x = \sqrt{2\lambda l}\left[\sqrt{n} - \sqrt{n-1}\right]$, this matches with pathfinder.

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